Annual Report - General Order No. R5-2007-0035
Reporting period 01/01/2023 to 12/31/2023.

DAIRY FACILITY INFORMATION

A. NAME OF DAIRY OR BUSINESS OPERATING THE DAIRY: Vitoria Dairy

Physical address of dairy:

Number and Street	City	County	Zip Code
14018 Crowslanding Road	Crowslanding	Stanislaus	95313

Street and nearest cross street (if no address):

Date facility was originally placed in operation: 02/15/1953

Regional Water Quality Control Board Basin Plan designation: San Joaquin River Basin

County Assessor Parcel Number(s) for dairy facility:

X057-X026-X007-XXXX X057-X026-X011-XXXX

B. OPERATORS

Vitoria, Jonathan

Operator name: Vitoria, Jonathan Telephone no.: (209) 652-9489 (Landline) (Cellular)

Mailing Address Number and Street	City	State	Zip Code
P.O. Box 327	Crows Landing	CA	95313

C. OWNERS

Brasil, Michael

Legal owner name: Brasil, Michael Telephone no.: (209) 595-2547 (Landline) (Cellular)

Mailing Address Number and Street	City	State	Zip Code
18246 First AVE	Stevinson	CA	95374

This owner is responsible for paying permit fees.

AVAILABLE NUTRIENTS

A. HERD INFORMATION

	Milk Cows	Dry Cows	Bred Heifers (15-24 mo.)	Heifers (7-14 mo. to breeding)	Calves (4-6 mo.)	Calves (0-3 mo.)
Number open confinement	0	0	299	272	180	0
Number under roof	1,015	100	0	0	0	0
Maximum number	1,015	100	299	272	180	0
Average number	1,015	100	299	272	180	0
Avg live weight (lbs)	1,400	1,450	1,000	750		

Predominant milk cow breed: Holstein

Average milk production: 68 *pounds per cow per day*

B. MANURE GENERATED

Total manure excreted by the herd: 10,632.41 *tons per reporting period*

Total nitrogen from manure: 117,429.32 *lbs per reporting period*

After ammonia losses (30% loss applied): 82,200.52 *lbs per reporting period*

Total phosphorus from manure: 18,284.66 *lbs per reporting period*

Total potassium from manure: 19,019.52 *lbs per reporting period*

Total salt from manure: 500,907.75 *lbs per reporting period*

C. PROCESS WASTEWATER GENERATED

Process wastewater generated: 23,800,000 *gallons*

Total nitrogen generated: 129,901.95 *lbs*

Total phosphorus generated: 16,685.32 *lbs*

Total potassium generated: 122,128.49 *lbs*

Total salt generated: 1,091,213.90 *lbs*

	23,800,000	*gallons applied*
+	0	*gallons exported*
-	0	*gallons imported*
=	23,800,000	*gallons generated*

D. FRESH WATER SOURCES

Source Description	Type
Domestic Well - Houses	Ground water
Domestic Well - Shop	Ground water
River water	Surface water

E. SUBSURFACE (TILE) DRAINAGE SOURCES

No subsurface (tile) drainage sources entered.

F. NUTRIENT IMPORTS

No dry manure nutrient imports entered.

No process wastewater nutrient imports entered.

No commercial or other nutrient imports entered.

G. NUTRIENT EXPORTS

No solid nutrient exports entered.

No liquid nutrient exports entered.

APPLICATION AREA

A. LIST OF LAND APPLICATION AREAS

Field name	Controlled acres	Cropable acres	Total harvests	Type of waste applied	Parcel number
V-1	15	15	2	process wastewater	X057-X026-X007-XXXX
V-2	25	25	2	process wastewater	X057-X026-X007-XXXX
V-3	26	26	2	process wastewater	X057-X026-X007-XXXX
V-4	11	11	2	process wastewater	X057-X026-X007-XXXX
V-5	33	33	2	process wastewater	X057-X026-X007-XXXX
V-6	55	55	2	process wastewater	X057-X026-X007-XXXX
V-7	35	35	2	process wastewater	X057-X026-X007-XXXX
V-8	30	30	2	process wastewater	X057-X026-X007-XXXX
V-9	10	10	2	process wastewater	X057-X026-X007-XXXX
Totals for areas that were used for application	240	240	18		
Totals for areas that were not used for application					
Land application area totals	240	240	18		

B. CROPS AND HARVESTS

V-1

Field name: V-1

11/03/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 15 Plant date: 11/03/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/10/2023	154.60 *ton*	Dry-weight		66.4	27,400.00	3,800.00	28,400.00		9.90

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	160.00	25.60	132.80	0.00
Total actual harvest content	10.31	189.77	26.32	196.70	685.68

V-1

06/26/2023: Corn, silage

Crop: Corn, silage Acres planted: 15 Plant date: 06/26/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
10/19/2023	436.50 *ton*	Dry-weight		68.4	13,100.00	2,400.00	16,900.00		5.80

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	30.00	240.00	45.00	198.00	0.00
Total actual harvest content	29.10	240.92	44.14	310.81	1,066.69

V-2

Field name: V-2

11/03/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 25 Plant date: 11/03/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/10/2023	249.90 *ton*	Dry-weight		66.4	27,400.00	3,800.00	28,400.00		9.90

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	160.00	25.60	132.80	0.00
Total actual harvest content	10.00	184.05	25.53	190.77	665.01

06/26/2023: Corn, silage

Crop: Corn, silage Acres planted: 25 Plant date: 06/26/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
10/19/2023	749.50 *ton*	Dry-weight		68.4	13,100.00	2,400.00	16,900.00		5.80

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	30.00	240.00	45.00	198.00	0.00
Total actual harvest content	29.98	248.21	45.47	320.21	1,098.95

V-3

Field name: V-3

11/03/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 26 Plant date: 11/03/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/10/2023	274.50 *ton*	Dry-weight		65.8	24,800.00	4,800.00	25,600.00		13.90

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	160.00	25.60	132.80	0.00
Total actual harvest content	10.56	179.09	34.66	184.87	1,003.78

06/26/2023: Corn, silage

Crop: Corn, silage Acres planted: 26 Plant date: 06/26/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
10/19/2023	775.30 *ton*	Dry-weight		67.5	10,800.00	1,700.00	16,500.00		6.30

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	30.00	240.00	45.00	198.00	0.00
Total actual harvest content	29.82	209.33	32.95	319.81	1,221.10

V-4

Field name: V-4

11/03/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 11 Plant date: 11/03/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/10/2023	110.60 *ton*	Dry-weight		65.8	24,800.00	4,800.00	25,600.00		13.90

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	160.00	25.60	132.80	0.00
Total actual harvest content	10.05	170.56	33.01	176.06	955.95

V-4

06/26/2023: Corn, silage

Crop: Corn, silage Acres planted: 11 Plant date: 06/26/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
10/19/2023	359.60 *ton*	Dry-weight		67.5	10,800.00	1,700.00	16,500.00		6.30

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	30.00	240.00	45.00	198.00	0.00
Total actual harvest content	32.69	229.49	36.12	350.61	1,338.69

V-5

Field name: V-5

11/03/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 33 Plant date: 11/03/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/10/2023	332.30 *ton*	Dry-weight		66.4	23,200.00	4,100.00	28,600.00		10.80

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	160.00	25.60	132.80	0.00
Total actual harvest content	10.07	156.99	27.74	193.53	730.82

06/26/2023: Corn, silage

Crop: Corn, silage Acres planted: 33 Plant date: 06/26/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
10/19/2023	987.60 *ton*	Dry-weight		68.0	13,800.00	2,400.00	16,400.00		6.50

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	30.00	240.00	45.00	198.00	0.00
Total actual harvest content	29.93	264.32	45.97	314.12	1,244.97

V-6

Field name: V-6

11/03/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 55 Plant date: 11/03/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/10/2023	601.20 *ton*	Dry-weight		66.5	26,700.00	4,600.00	27,200.00		13.70

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	160.00	25.60	132.80	0.00
Total actual harvest content	10.93	195.54	33.69	199.20	1,003.35

06/26/2023: Corn, silage

Crop: Corn, silage Acres planted: 55 Plant date: 06/26/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
10/19/2023	1,612.30 *ton*	Dry-weight		66.3	13,800.00	2,200.00	15,000.00		5.60

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	30.00	240.00	45.00	198.00	0.00
Total actual harvest content	29.31	272.66	43.47	296.37	1,106.45

V-7

Field name: V-7

11/03/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 35 Plant date: 11/03/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/10/2023	359.60 *ton*	Dry-weight		65.2	23,700.00	4,300.00	26,700.00		11.80

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	160.00	25.60	132.80	0.00
Total actual harvest content	10.27	169.48	30.75	190.93	843.81

V-7

06/26/2023: Corn, silage

Crop: Corn, silage Acres planted: 35 Plant date: 06/26/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
10/19/2023	1,049.60 *ton*	Dry-weight		62.3	13,900.00	2,800.00	15,000.00		5.80

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	30.00	240.00	45.00	198.00	0.00
Total actual harvest content	29.99	314.30	63.31	339.17	1,311.46

V-8

Field name: V-8

11/03/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 30 Plant date: 11/03/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/10/2023	285.60 *ton*	Dry-weight		66.0	27,000.00	4,400.00	28,200.00		10.50

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	160.00	25.60	132.80	0.00
Total actual harvest content	9.52	174.79	28.48	182.56	679.73

06/26/2023: Corn, silage

Crop: Corn, silage Acres planted: 30 Plant date: 06/26/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
10/19/2023	801.30 *ton*	Dry-weight		61.4	12,100.00	2,000.00	15,200.00		5.90

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	30.00	240.00	45.00	198.00	0.00
Total actual harvest content	26.71	249.50	41.24	313.43	1,216.59

V-9

Field name: V-9

11/03/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 10 Plant date: 11/03/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/10/2023	90.60 *ton*	Dry-weight		66.0	27,000.00	4,400.00	28,200.00		10.50

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	160.00	25.60	132.80	0.00
Total actual harvest content	9.06	166.34	27.11	173.73	646.88

06/26/2023: Corn, silage

Crop: Corn, silage Acres planted: 10 Plant date: 06/26/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
10/19/2023	271.40 *ton*	Dry-weight		61.4	12,100.00	2,000.00	15,200.00		5.90

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	30.00	240.00	45.00	198.00	0.00
Total actual harvest content	27.14	253.52	41.90	318.47	1,236.17

NUTRIENT BUDGET

A. LAND APPLICATIONS

V-1 - 11/03/2022: Oats, silage-soft dough

Field name: V-1

Crop: Oats, silage-soft dough — Plant date: 11/03/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
03/15/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	139.95	16.01	107.86	1,014.81	370,000.00 *gal*
River water	Surface water	0.13	0.00	0.00	53.00	2,323,575.00 *gal*
Application event totals		140.08	16.01	107.86	1,067.81	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/13/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	80.64	10.60	80.57	812.91	260,000.00 *gal*
River water	Surface water	0.12	0.00	0.00	49.93	2,188,875.00 *gal*
Application event totals		80.76	10.60	80.57	862.84	

V-1 - 06/26/2023: Corn, silage

Field name: V-1

Crop: Corn, silage — Plant date: 06/26/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/19/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
River water	Surface water	0.15	0.00	0.00	60.68	2,660,325.00 *gal*
Application event totals		0.15	0.00	0.00	60.68	

V-1 - 06/26/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/20/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
River water	Surface water	0.14	0.00	0.00	58.38	2,559,300.00 *gal*
Application event totals		0.14	0.00	0.00	58.38	

08/05/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	83.22	9.52	64.13	603.40	220,000.00 *gal*
River water	Surface water	0.12	0.00	0.00	48.39	2,121,525.00 *gal*
Application event totals		83.33	9.52	64.13	651.79	

08/19/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	71.87	8.22	55.39	521.12	190,000.00 *gal*
River water	Surface water	0.12	0.00	0.00	50.70	2,222,550.00 *gal*
Application event totals		71.99	8.22	55.39	571.81	

09/02/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	75.65	8.66	58.30	548.54	200,000.00 *gal*
River water	Surface water	0.12	0.00	0.00	48.39	2,121,525.00 *gal*
Application event totals		75.77	8.66	58.30	596.94	

09/16/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	87.00	9.96	67.05	630.83	230,000.00 *gal*
River water	Surface water	0.12	0.00	0.00	49.16	2,155,200.00 *gal*
Application event totals		87.12	9.96	67.05	679.99	

V-1 - 06/26/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/30/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
River water	Surface water	0.12	0.00	0.00	47.62	2,087,850.00 *gal*
Application event totals		0.12	0.00	0.00	47.62	

V-2 - 11/03/2022: Oats, silage-soft dough

Field name: V-2

Crop: Oats, silage-soft dough

Plant date: 11/03/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
03/15/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	102.35	13.46	102.26	1,031.78	550,000.00 *gal*
River water	Surface water	0.12	0.00	0.00	50.70	3,704,250.00 *gal*
Application event totals		102.48	13.46	102.26	1,082.47	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/13/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	120.28	13.76	92.70	872.19	530,000.00 *gal*
River water	Surface water	0.12	0.00	0.00	50.23	3,670,575.00 *gal*
Application event totals		120.41	13.76	92.70	922.42	

V-2 - 06/26/2023: Corn, silage

Field name: V-2

Crop: Corn, silage

Plant date: 06/26/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following

V-2 - 06/26/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/19/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
River water	Surface water	0.13	0.00	0.00	51.62	3,771,600.00 *gal*
Application event totals		0.13	0.00	0.00	51.62	

07/20/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
River water	Surface water	0.12	0.00	0.00	50.70	3,704,250.00 *gal*
Application event totals		0.12	0.00	0.00	50.70	

08/05/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	93.26	12.97	98.16	770.68	390,000.00 *gal*
River water	Surface water	0.12	0.00	0.00	51.16	3,737,925.00 *gal*
Application event totals		93.39	12.97	98.16	821.83	

08/19/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	74.44	9.79	74.37	750.38	400,000.00 *gal*
River water	Surface water	0.12	0.00	0.00	51.16	3,737,925.00 *gal*
Application event totals		74.56	9.79	74.37	801.54	

09/02/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	83.97	9.61	64.72	608.88	370,000.00 *gal*
River water	Surface water	0.12	0.00	0.00	50.70	3,704,250.00 *gal*
Application event totals		84.10	9.61	64.72	659.58	

V-2 - 06/26/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/16/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	83.97	9.61	64.72	608.88	370,000.00 *gal*
River water	Surface water	0.12	0.00	0.00	51.16	3,737,925.00 *gal*
Application event totals		84.10	9.61	64.72	660.04	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/30/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
River water	Surface water	0.13	0.00	0.00	51.62	3,771,600.00 *gal*
Application event totals		0.13	0.00	0.00	51.62	

V-3 - 11/03/2022: Oats, silage-soft dough

Field name: V-3

Crop: Oats, silage-soft dough Plant date: 11/03/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
03/15/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	122.20	13.98	94.18	886.11	560,000.00 *gal*
River water	Surface water	0.12	0.00	0.00	48.75	3,704,250.00 *gal*
Application event totals		122.32	13.98	94.18	934.86	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/13/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	98.41	12.94	98.33	992.09	550,000.00 *gal*
River water	Surface water	0.12	0.00	0.00	49.63	3,771,600.00 *gal*
Application event totals		98.54	12.94	98.33	1,041.72	

V-3 - 06/26/2023: Corn, silage

V-3 - 06/26/2023: Corn, silage

Field name: V-3

Crop: Corn, silage

Plant date: 06/26/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/19/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
River water	Surface water	0.12	0.00	0.00	50.96	3,872,625.00 *gal*
Application event totals		0.12	0.00	0.00	50.96	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/20/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
River water	Surface water	0.12	0.00	0.00	50.08	3,805,275.00 *gal*
Application event totals		0.12	0.00	0.00	50.08	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/05/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	69.79	9.18	69.72	703.48	390,000.00 *gal*
River water	Surface water	0.12	0.00	0.00	48.75	3,704,250.00 *gal*
Application event totals		69.90	9.18	69.72	752.23	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/19/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	73.36	9.65	73.30	739.56	410,000.00 *gal*
River water	Surface water	0.12	0.00	0.00	50.96	3,872,625.00 *gal*
Application event totals		73.49	9.65	73.30	790.52	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/02/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	63.28	7.24	48.77	458.88	290,000.00 *gal*
River water	Surface water	0.12	0.00	0.00	50.08	3,805,275.00 *gal*
Application event totals		63.41	7.24	48.77	508.95	

V-3 - 06/26/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/16/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	71.28	9.91	75.02	589.03	310,000.00 *gal*
River water	Surface water	0.12	0.00	0.00	48.75	3,704,250.00 *gal*
Application event totals		71.40	9.91	75.02	637.77	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/30/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
River water	Surface water	0.12	0.00	0.00	50.08	3,805,275.00 *gal*
Application event totals		0.12	0.00	0.00	50.08	

V-4 - 11/03/2022: Oats, silage-soft dough

Field name: V-4

Crop: Oats, silage-soft dough

Plant date: 11/03/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
03/15/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	139.27	15.94	107.33	1,009.82	270,000.00 *gal*
River water	Surface water	0.14	0.00	0.00	55.51	1,784,775.00 *gal*
Application event totals		139.40	15.94	107.33	1,065.33	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/13/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	77.37	8.85	59.63	561.01	150,000.00 *gal*
River water	Surface water	0.13	0.00	0.00	53.42	1,717,425.00 *gal*
Application event totals		77.50	8.85	59.63	614.43	

V-4 - 06/26/2023: Corn, silage

V-4 - 06/26/2023: Corn, silage

Field name: V-4

Crop: Corn, silage

Plant date: 06/26/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/19/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
River water	Surface water	0.14	0.00	0.00	57.61	1,852,125.00 *gal*
Application event totals		0.14	0.00	0.00	57.61	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/20/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
River water	Surface water	0.13	0.00	0.00	54.47	1,751,100.00 *gal*
Application event totals		0.13	0.00	0.00	54.47	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/05/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	80.36	10.57	80.29	810.07	190,000.00 *gal*
River water	Surface water	0.14	0.00	0.00	57.61	1,852,125.00 *gal*
Application event totals		80.50	10.57	80.29	867.68	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/19/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	71.90	9.45	71.84	724.80	170,000.00 *gal*
River water	Surface water	0.14	0.00	0.00	55.51	1,784,775.00 *gal*
Application event totals		72.04	9.45	71.84	780.31	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/02/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	59.21	7.79	59.16	596.90	140,000.00 *gal*
River water	Surface water	0.13	0.00	0.00	53.42	1,717,425.00 *gal*
Application event totals		59.34	7.79	59.16	650.31	

V-4 - 06/26/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/16/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	98.00	11.21	75.53	710.61	190,000.00 *gal*
River water	Surface water	0.13	0.00	0.00	52.37	1,683,750.00 *gal*
Application event totals		98.13	11.21	75.53	762.99	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/30/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
River water	Surface water	0.13	0.00	0.00	53.42	1,717,425.00 *gal*
Application event totals		0.13	0.00	0.00	53.42	

V-5 - 11/03/2022: Oats, silage-soft dough

Field name: V-5

Crop: Oats, silage-soft dough

Plant date: 11/03/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
03/15/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	108.32	12.39	83.48	785.42	630,000.00 *gal*
River water	Surface water	0.13	0.00	0.00	52.02	5,017,575.00 *gal*
Application event totals		108.44	12.39	83.48	837.44	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/13/2023	Pipeline	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	97.83	13.60	102.96	808.40	540,000.00 *gal*
River water	Surface water	0.13	0.00	0.00	51.32	4,950,225.00 *gal*
Application event totals		97.95	13.60	102.96	859.73	

V-5 - 06/26/2023: Corn, silage

V-5 - 06/26/2023: Corn, silage

Field name: V-5

Crop: Corn, silage

Plant date: 06/26/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/19/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
River water	Surface water	0.13	0.00	0.00	52.72	5,084,925.00 *gal*
Application event totals		0.13	0.00	0.00	52.72	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/20/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
River water	Surface water	0.12	0.00	0.00	50.97	4,916,550.00 *gal*
Application event totals		0.12	0.00	0.00	50.97	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/05/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	99.64	13.85	104.87	823.37	550,000.00 *gal*
River water	Surface water	0.12	0.00	0.00	50.97	4,916,550.00 *gal*
Application event totals		99.76	13.85	104.87	874.35	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/19/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	103.26	14.36	108.68	853.31	570,000.00 *gal*
River water	Surface water	0.12	0.00	0.00	50.63	4,882,875.00 *gal*
Application event totals		103.39	14.36	108.68	903.94	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/02/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	77.37	8.85	59.63	561.01	450,000.00 *gal*
River water	Surface water	0.12	0.00	0.00	50.63	4,882,875.00 *gal*
Application event totals		77.49	8.85	59.63	611.64	

V-5 - 06/26/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/16/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	68.77	7.87	53.00	498.68	400,000.00 *gal*
River water	Surface water	0.13	0.00	0.00	52.72	5,084,925.00 *gal*
Application event totals		68.90	7.87	53.00	551.40	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/30/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
River water	Surface water	0.12	0.00	0.00	50.63	4,882,875.00 *gal*
Application event totals		0.12	0.00	0.00	50.63	

V-6 - 11/03/2022: Oats, silage-soft dough

Field name: V-6

Crop: Oats, silage-soft dough

Plant date: 11/03/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
03/15/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	100.66	13.23	100.57	1,014.72	1,190,000.00 *gal*
River water	Surface water	0.12	0.00	0.00	50.49	8,115,675.00 *gal*
Application event totals		100.78	13.23	100.57	1,065.21	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/13/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	106.52	14.81	112.11	880.26	980,000.00 *gal*
River water	Surface water	0.13	0.00	0.00	51.74	8,317,725.00 *gal*
Application event totals		106.65	14.81	112.11	932.00	

V-6 - 06/26/2023: Corn, silage

V-6 - 06/26/2023: Corn, silage

Field name: V-6

Crop: Corn, silage

Plant date: 06/26/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/19/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
River water	Surface water	0.12	0.00	0.00	50.70	8,149,350.00 *gal*
Application event totals		0.12	0.00	0.00	50.70	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/20/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
River water	Surface water	0.12	0.00	0.00	50.91	8,183,025.00 *gal*
Application event totals		0.12	0.00	0.00	50.91	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/05/2023	Pipeline	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	92.39	12.85	97.24	763.49	850,000.00 *gal*
River water	Surface water	0.13	0.00	0.00	51.74	8,317,725.00 *gal*
Application event totals		92.52	12.85	97.24	815.23	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/19/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	91.31	12.69	96.10	754.51	840,000.00 *gal*
River water	Surface water	0.13	0.00	0.00	51.32	8,250,375.00 *gal*
Application event totals		91.43	12.69	96.10	805.83	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/02/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	81.50	9.33	62.81	590.93	790,000.00 *gal*
River water	Surface water	0.12	0.00	0.00	51.11	8,216,700.00 *gal*
Application event totals		81.62	9.33	62.81	642.05	

V-6 - 06/26/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/16/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	73.91	10.28	77.79	610.79	680,000.00 *gal*
River water	Surface water	0.12	0.00	0.00	51.11	8,216,700.00 *gal*
Application event totals		74.04	10.28	77.79	661.91	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/30/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
River water	Surface water	0.12	0.00	0.00	50.49	8,115,675.00 *gal*
Application event totals		0.12	0.00	0.00	50.49	

V-7 - 11/03/2022: Oats, silage-soft dough

Field name: V-7

Crop: Oats, silage-soft dough Plant date: 11/03/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
03/15/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	107.61	14.96	113.26	889.24	630,000.00 *gal*
River water	Surface water	0.12	0.00	0.00	49.71	5,084,925.00 *gal*
Application event totals		107.73	14.96	113.26	938.95	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/13/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	114.44	15.91	120.45	945.70	670,000.00 *gal*
River water	Surface water	0.13	0.00	0.00	53.00	5,421,675.00 *gal*
Application event totals		114.57	15.91	120.45	998.70	

V-7 - 06/26/2023: Corn, silage

V-7 - 06/26/2023: Corn, silage

Field name: V-7

Crop: Corn, silage

Plant date: 06/26/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/19/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
River water	Surface water	0.13	0.00	0.00	52.34	5,354,325.00 *gal*
Application event totals		0.13	0.00	0.00	52.34	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/20/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
River water	Surface water	0.13	0.00	0.00	53.00	5,421,675.00 *gal*
Application event totals		0.13	0.00	0.00	53.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/05/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	65.13	8.56	65.07	656.58	490,000.00 *gal*
River water	Surface water	0.13	0.00	0.00	52.67	5,388,000.00 *gal*
Application event totals		65.26	8.56	65.07	709.26	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/19/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	67.79	8.91	67.73	683.38	510,000.00 *gal*
River water	Surface water	0.13	0.00	0.00	52.01	5,320,650.00 *gal*
Application event totals		67.92	8.91	67.73	735.40	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/02/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	86.40	11.36	86.32	870.98	650,000.00 *gal*
River water	Surface water	0.13	0.00	0.00	53.66	5,489,025.00 *gal*
Application event totals		86.53	11.36	86.32	924.64	

V-7 - 06/26/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/16/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	85.07	11.19	85.00	857.58	640,000.00 *gal*
River water	Surface water	0.13	0.00	0.00	52.34	5,354,325.00 *gal*
Application event totals		85.20	11.19	85.00	909.92	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/30/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
River water	Surface water	0.13	0.00	0.00	51.68	5,286,975.00 *gal*
Application event totals		0.13	0.00	0.00	51.68	

V-8 - 11/03/2022: Oats, silage-soft dough

Field name: V-8

Crop: Oats, silage-soft dough

Plant date: 11/03/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
03/15/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	125.55	17.45	132.13	1,037.45	630,000.00 *gal*
River water	Surface water	0.12	0.00	0.00	47.24	4,142,025.00 *gal*
Application event totals		125.66	17.45	132.13	1,084.69	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/13/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	96.15	12.64	96.06	969.24	620,000.00 *gal*
River water	Surface water	0.11	0.00	0.00	46.86	4,108,350.00 *gal*
Application event totals		96.26	12.64	96.06	1,016.10	

V-8 - 06/26/2023: Corn, silage

V-8 - 06/26/2023: Corn, silage

Field name: V-8

Crop: Corn, silage

Plant date: 06/26/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/19/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
River water	Surface water	0.13	0.00	0.00	52.23	4,579,800.00 *gal*
Application event totals		0.13	0.00	0.00	52.23	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/20/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
River water	Surface water	0.13	0.00	0.00	53.38	4,680,825.00 *gal*
Application event totals		0.13	0.00	0.00	53.38	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/05/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	83.70	11.64	88.09	691.63	420,000.00 *gal*
River water	Surface water	0.13	0.00	0.00	52.62	4,613,475.00 *gal*
Application event totals		83.83	11.64	88.09	744.25	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/19/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	87.00	9.96	67.05	630.83	460,000.00 *gal*
River water	Surface water	0.13	0.00	0.00	52.62	4,613,475.00 *gal*
Application event totals		87.13	9.96	67.05	683.44	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/02/2023	Pipeline	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	96.45	11.04	74.34	699.39	510,000.00 *gal*
River water	Surface water	0.12	0.00	0.00	51.08	4,478,775.00 *gal*
Application event totals		96.58	11.04	74.34	750.47	

V-8 - 06/26/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/16/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	73.76	8.44	56.85	534.83	390,000.00 *gal*
River water	Surface water	0.12	0.00	0.00	51.08	4,478,775.00 *gal*
Application event totals		73.88	8.44	56.85	585.91	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/30/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
River water	Surface water	0.13	0.00	0.00	51.85	4,546,125.00 *gal*
Application event totals		0.13	0.00	0.00	51.85	

V-9 - 11/03/2022: Oats, silage-soft dough

Field name: V-9

Crop: Oats, silage-soft dough Plant date: 11/03/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
03/15/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	130.50	14.93	100.57	946.24	230,000.00 *gal*
River water	Surface water	0.11	0.00	0.00	46.09	1,347,000.00 *gal*
Application event totals		130.61	14.93	100.57	992.33	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/13/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	83.70	11.64	88.09	691.63	140,000.00 *gal*
River water	Surface water	0.11	0.00	0.00	44.93	1,313,325.00 *gal*
Application event totals		83.81	11.64	88.09	736.57	

V-9 - 06/26/2023: Corn, silage

V-9 - 06/26/2023: Corn, silage

Field name: V-9

Crop: Corn, silage

Plant date: 06/26/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/19/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
River water	Surface water	0.15	0.00	0.00	63.37	1,852,125.00 *gal*
Application event totals		0.15	0.00	0.00	63.37	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/20/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
River water	Surface water	0.16	0.00	0.00	66.83	1,953,150.00 *gal*
Application event totals		0.16	0.00	0.00	66.83	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/05/2023	Pipeline	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	107.00	14.07	106.91	1,078.67	230,000.00 *gal*
River water	Surface water	0.14	0.00	0.00	58.76	1,717,425.00 *gal*
Application event totals		107.15	14.07	106.91	1,137.44	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/19/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	90.78	10.39	69.96	658.25	160,000.00 *gal*
River water	Surface water	0.10	0.00	0.00	42.63	1,245,975.00 *gal*
Application event totals		90.88	10.39	69.96	700.88	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/02/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	77.72	10.81	81.80	642.23	130,000.00 *gal*
River water	Surface water	0.11	0.00	0.00	44.93	1,313,325.00 *gal*
Application event totals		77.83	10.81	81.80	687.17	

V-9 - 06/26/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/16/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	55.83	7.34	55.78	562.79	120,000.00 *gal*
River water	Surface water	0.11	0.00	0.00	43.78	1,279,650.00 *gal*
Application event totals		55.93	7.34	55.78	606.57	

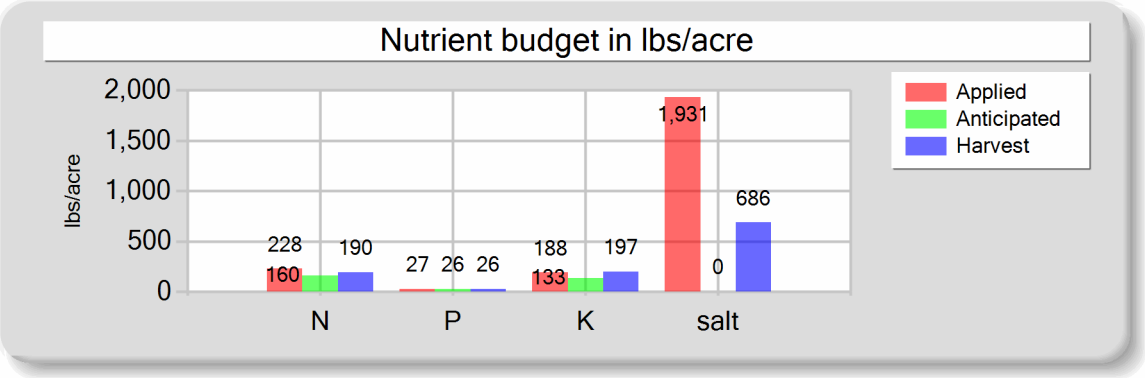
Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/30/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
River water	Surface water	0.12	0.00	0.00	49.54	1,448,025.00 *gal*
Application event totals		0.12	0.00	0.00	49.54	

B. NUTRIENT BUDGET

V-1 - 11/03/2022: Oats, silage-soft dough

Field name: V-1 Crop: Oats, silage-soft dough Plant date: 11/03/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	220.59	26.62	188.43	1,827.72
Fresh water	0.25	0.00	0.00	102.93
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	227.84	26.62	188.43	1,930.65
Anticipated crop nutrient removal	160.00	25.60	132.80	0.00
Actual crop nutrient removal	189.77	26.32	196.70	685.68
Nutrient balance	38.07	0.30	-8.27	1,244.97
Applied to removed ratio	1.20	1.01	0.96	2.82

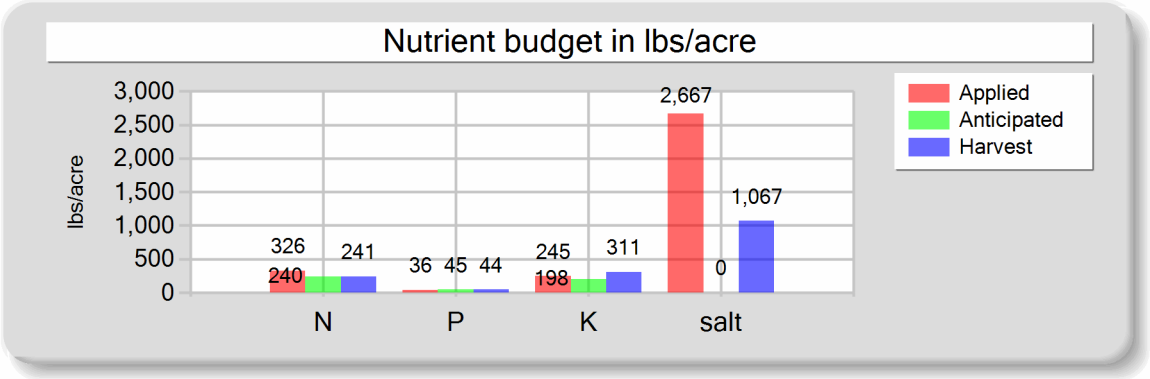
Fresh water applied	
4,512,450.00	*gallons*
166.18	*acre-inches*
11.08	*inches/acre*

Process wastewater applied	
630,000.00	*gallons*
23.20	*acre-inches*
1.55	*inches/acre*

Total harvests for the crop	
1	*harvests*

V-1 - 06/26/2023: Corn, silage

Field name: V-1 Crop: Corn, silage Plant date: 06/26/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	317.73	36.36	244.88	2,303.89
Fresh water	0.89	0.00	0.00	363.32
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	325.62	36.36	244.88	2,667.21
Anticipated crop nutrient removal	240.00	45.00	198.00	0.00
Actual crop nutrient removal	240.92	44.14	310.81	1,066.69
Nutrient balance	84.69	-7.78	-65.94	1,600.52
Applied to removed ratio	1.35	0.82	0.79	2.50

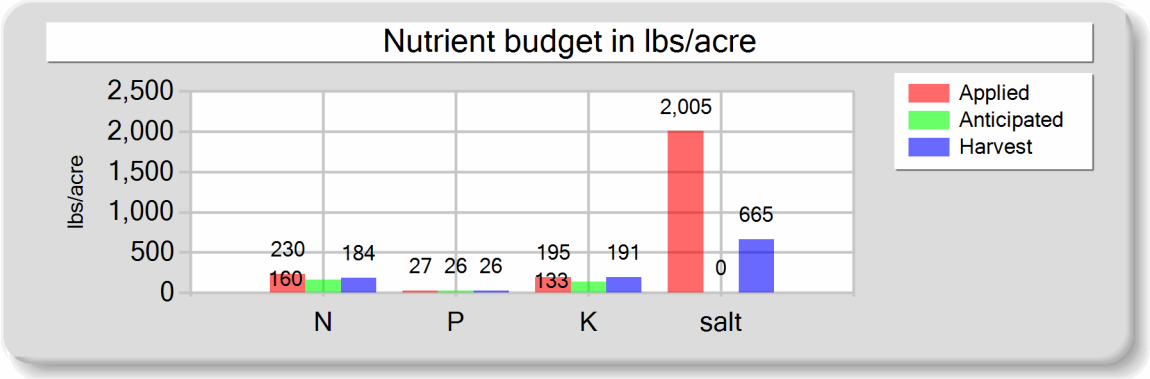
Fresh water applied	
15,928,275.00	*gallons*
586.58	*acre-inches*
39.11	*inches/acre*

Process wastewater applied	
840,000.00	*gallons*
30.93	*acre-inches*
2.06	*inches/acre*

Total harvests for the crop	
1	*harvests*

V-2 - 11/03/2022: Oats, silage-soft dough

Field name: V-2 Crop: Oats, silage-soft dough Plant date: 11/03/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	222.64	27.22	194.96	1,903.96
Fresh water	0.25	0.00	0.00	100.93
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	229.88	27.22	194.96	2,004.89
Anticipated crop nutrient removal	160.00	25.60	132.80	0.00
Actual crop nutrient removal	184.05	25.53	190.77	665.01
Nutrient balance	45.83	1.70	4.19	1,339.88
Applied to removed ratio	1.25	1.07	1.02	3.01

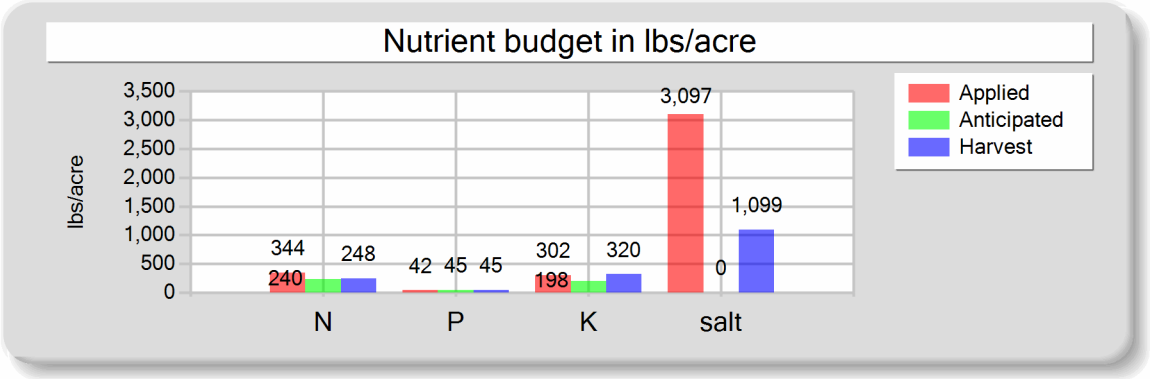
Fresh water applied	
7,374,825.00	*gallons*
271.59	*acre-inches*
10.86	*inches/acre*

Process wastewater applied	
1,080,000.00	*gallons*
39.77	*acre-inches*
1.59	*inches/acre*

Total harvests for the crop	
1	*harvests*

V-2 - 06/26/2023: Corn, silage

Field name: V-2 Crop: Corn, silage Plant date: 06/26/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	335.64	41.97	301.96	2,738.83
Fresh water	0.87	0.00	0.00	358.10
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	343.52	41.97	301.96	3,096.92
Anticipated crop nutrient removal	240.00	45.00	198.00	0.00
Actual crop nutrient removal	248.21	45.47	320.21	1,098.95
Nutrient balance	95.31	-3.50	-18.25	1,997.98
Applied to removed ratio	1.38	0.92	0.94	2.82

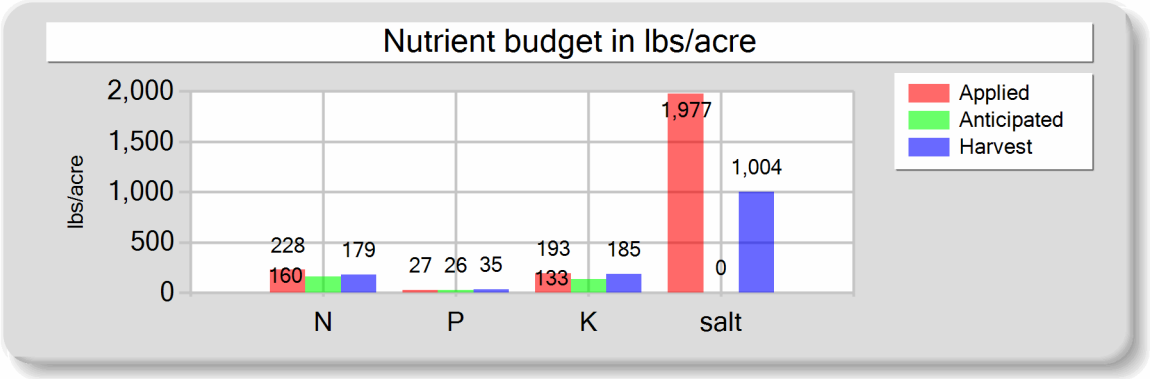
Fresh water applied	
26,165,475.00	*gallons*
963.59	*acre-inches*
38.54	*inches/acre*

Process wastewater applied	
1,530,000.00	*gallons*
56.34	*acre-inches*
2.25	*inches/acre*

Total harvests for the crop	
1	*harvests*

V-3 - 11/03/2022: Oats, silage-soft dough

Field name: V-3 Crop: Oats, silage-soft dough Plant date: 11/03/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	220.62	26.92	192.51	1,878.20
Fresh water	0.24	0.00	0.00	98.38
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	227.86	26.92	192.51	1,976.58
Anticipated crop nutrient removal	160.00	25.60	132.80	0.00
Actual crop nutrient removal	179.09	34.66	184.87	1,003.78
Nutrient balance	48.77	-7.74	7.64	972.80
Applied to removed ratio	1.27	0.78	1.04	1.97

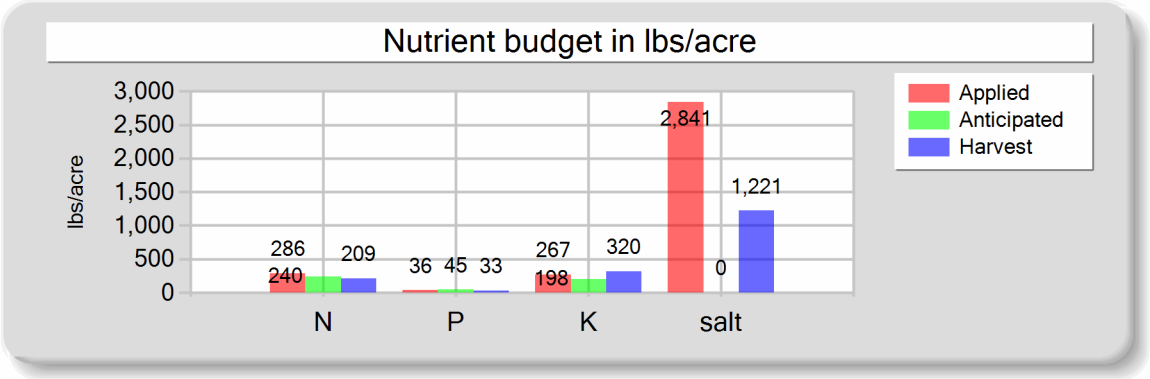
Fresh water applied	
7,475,850.00	*gallons*
275.31	*acre-inches*
10.59	*inches/acre*

Process wastewater applied	
1,110,000.00	*gallons*
40.88	*acre-inches*
1.57	*inches/acre*

Total harvests for the crop	
1	*harvests*

V-3 - 06/26/2023: Corn, silage

Field name: V-3 Crop: Corn, silage Plant date: 06/26/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	277.71	35.97	266.82	2,490.95
Fresh water	0.85	0.00	0.00	349.64
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	285.57	35.97	266.82	2,840.59
Anticipated crop nutrient removal	240.00	45.00	198.00	0.00
Actual crop nutrient removal	209.33	32.95	319.81	1,221.10
Nutrient balance	76.24	3.02	-53.00	1,619.49
Applied to removed ratio	1.36	1.09	0.83	2.33

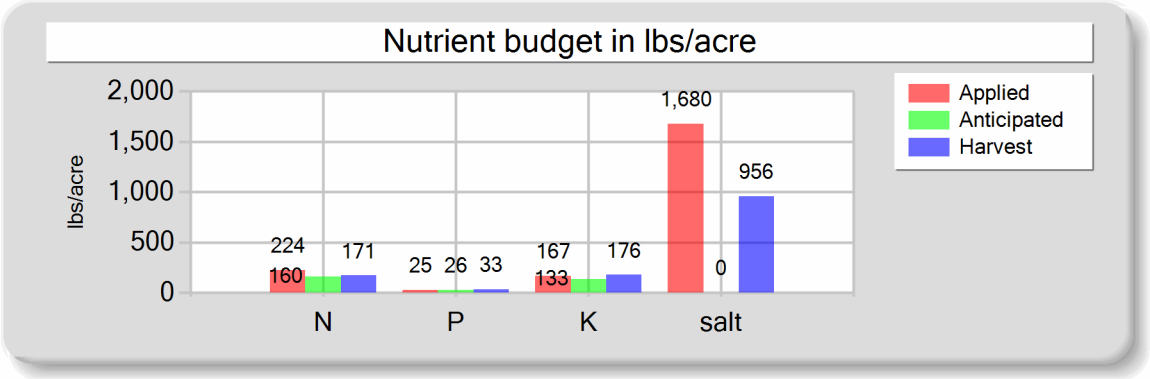
Fresh water applied	
26,569,575.00	*gallons*
978.47	*acre-inches*
37.63	*inches/acre*

Process wastewater applied	
1,400,000.00	*gallons*
51.56	*acre-inches*
1.98	*inches/acre*

Total harvests for the crop	
1	*harvests*

V-4 - 11/03/2022: Oats, silage-soft dough

Field name: V-4 Crop: Oats, silage-soft dough Plant date: 11/03/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	216.63	24.79	166.96	1,570.83
Fresh water	0.27	0.00	0.00	108.93
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	223.90	24.79	166.96	1,679.77
Anticipated crop nutrient removal	160.00	25.60	132.80	0.00
Actual crop nutrient removal	170.56	33.01	176.06	955.95
Nutrient balance	53.34	-8.22	-9.10	723.82
Applied to removed ratio	1.31	0.75	0.95	1.76

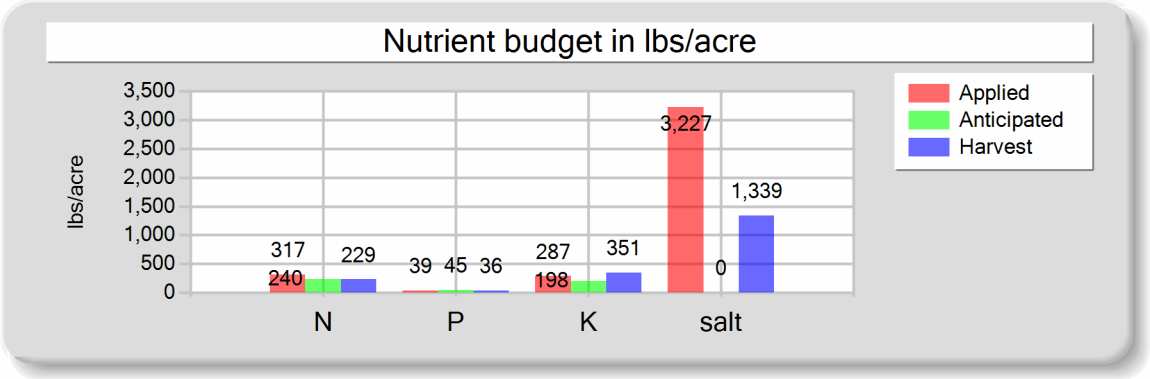
Fresh water applied	
3,502,200.00	*gallons*
128.97	*acre-inches*
11.72	*inches/acre*

Process wastewater applied	
420,000.00	*gallons*
15.47	*acre-inches*
1.41	*inches/acre*

Total harvests for the crop	
1	*harvests*

V-4 - 06/26/2023: Corn, silage

Field name: V-4 Crop: Corn, silage Plant date: 06/26/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	309.47	39.02	286.81	2,842.38
Fresh water	0.94	0.00	0.00	384.41
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	317.41	39.02	286.81	3,226.79
Anticipated crop nutrient removal	240.00	45.00	198.00	0.00
Actual crop nutrient removal	229.49	36.12	350.61	1,338.69
Nutrient balance	87.92	2.89	-63.80	1,888.10
Applied to removed ratio	1.38	1.08	0.82	2.41

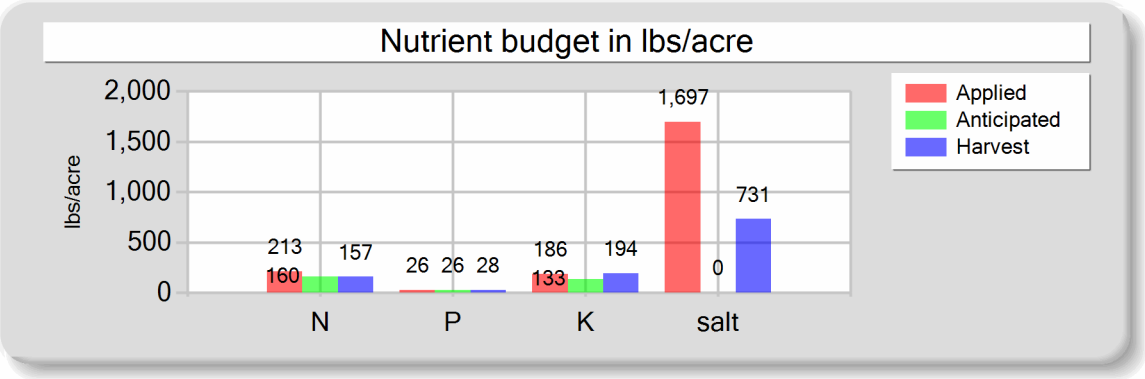
Fresh water applied	
12,358,725.00	*gallons*
455.13	*acre-inches*
41.38	*inches/acre*

Process wastewater applied	
690,000.00	*gallons*
25.41	*acre-inches*
2.31	*inches/acre*

Total harvests for the crop	
1	*harvests*

V-5 - 11/03/2022: Oats, silage-soft dough

Field name: V-5 Crop: Oats, silage-soft dough Plant date: 11/03/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	206.15	26.00	186.44	1,593.82
Fresh water	0.25	0.00	0.00	103.35
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	213.40	26.00	186.44	1,697.17
Anticipated crop nutrient removal	160.00	25.60	132.80	0.00
Actual crop nutrient removal	156.99	27.74	193.53	730.82
Nutrient balance	56.41	-1.75	-7.09	966.35
Applied to removed ratio	1.36	0.94	0.96	2.32

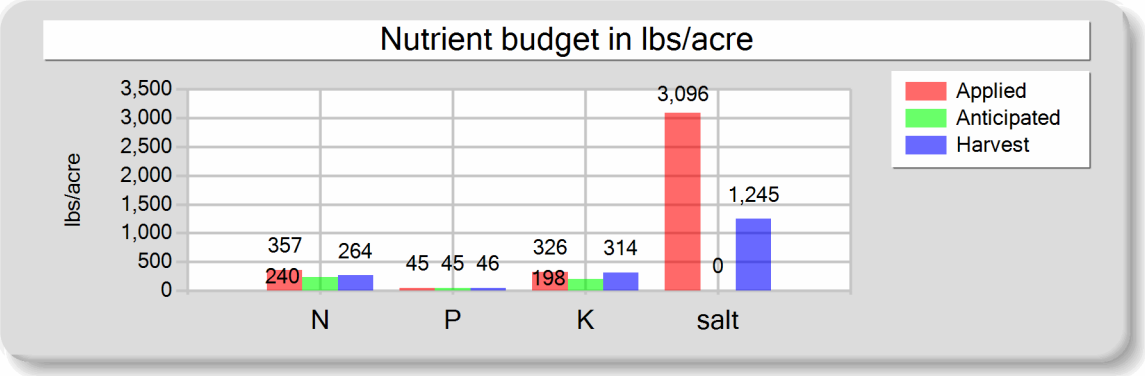
Fresh water applied	
9,967,800.00	*gallons*
367.08	*acre-inches*
11.12	*inches/acre*

Process wastewater applied	
1,170,000.00	*gallons*
43.09	*acre-inches*
1.31	*inches/acre*

Total harvests for the crop	
1	*harvests*

V-5 - 06/26/2023: Corn, silage

Field name: V-5 Crop: Corn, silage Plant date: 06/26/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	349.04	44.93	326.18	2,736.38
Fresh water	0.88	0.00	0.00	359.27
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	356.92	44.93	326.18	3,095.64
Anticipated crop nutrient removal	240.00	45.00	198.00	0.00
Actual crop nutrient removal	264.32	45.97	314.12	1,244.97
Nutrient balance	92.60	-1.04	12.07	1,850.67
Applied to removed ratio	1.35	0.98	1.04	2.49

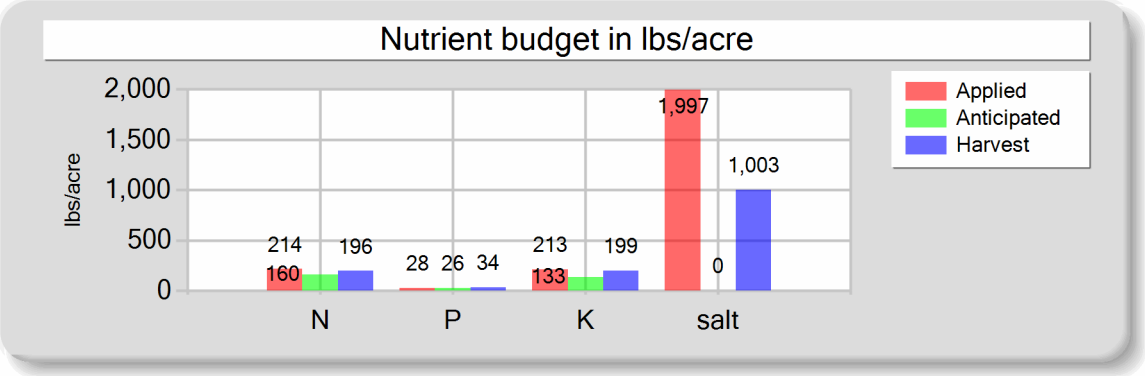
Fresh water applied	
34,651,575.00	*gallons*
1,276.10	*acre-inches*
38.67	*inches/acre*

Process wastewater applied	
1,970,000.00	*gallons*
72.55	*acre-inches*
2.20	*inches/acre*

Total harvests for the crop	
1	*harvests*

V-6 - 11/03/2022: Oats, silage-soft dough

Field name: V-6 Crop: Oats, silage-soft dough Plant date: 11/03/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	207.18	28.04	212.68	1,894.98
Fresh water	0.25	0.00	0.00	102.23
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	214.43	28.04	212.68	1,997.21
Anticipated crop nutrient removal	160.00	25.60	132.80	0.00
Actual crop nutrient removal	195.54	33.69	199.20	1,003.35
Nutrient balance	18.89	-5.64	13.48	993.86
Applied to removed ratio	1.10	0.83	1.07	1.99

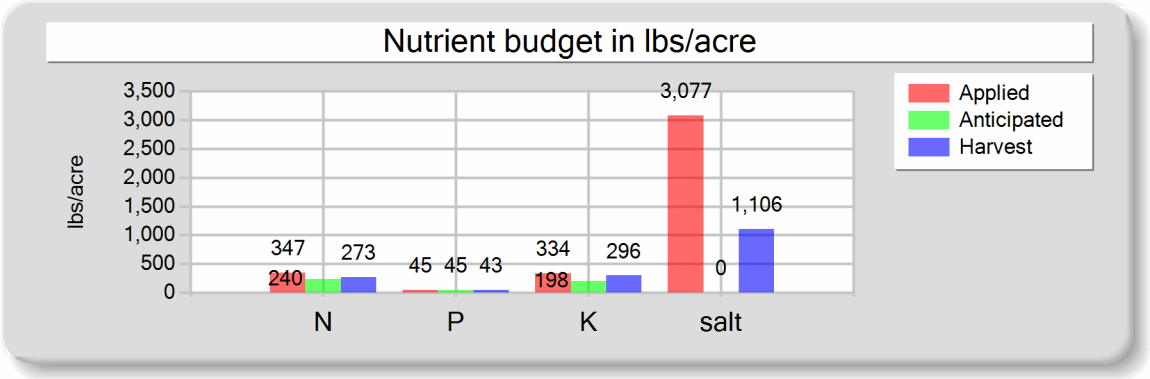
Fresh water applied	
16,433,400.00	*gallons*
605.19	*acre-inches*
11.00	*inches/acre*

Process wastewater applied	
2,170,000.00	*gallons*
79.91	*acre-inches*
1.45	*inches/acre*

Total harvests for the crop	
1	*harvests*

V-6 - 06/26/2023: Corn, silage

Field name: V-6 Crop: Corn, silage Plant date: 06/26/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	339.11	45.14	333.94	2,719.73
Fresh water	0.87	0.00	0.00	357.38
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	346.98	45.14	333.94	3,077.11
Anticipated crop nutrient removal	240.00	45.00	198.00	0.00
Actual crop nutrient removal	272.66	43.47	296.37	1,106.45
Nutrient balance	74.32	1.67	37.57	1,970.66
Applied to removed ratio	1.27	1.04	1.13	2.78

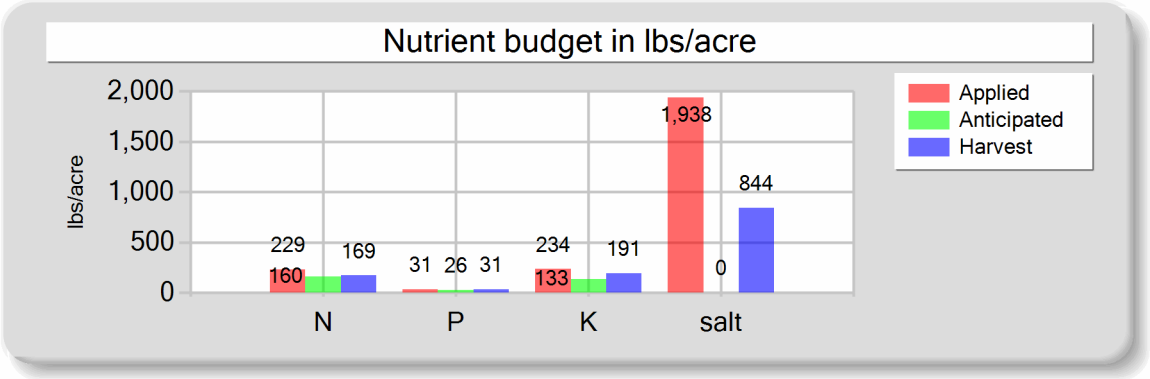
Fresh water applied	
57,449,550.00	*gallons*
2,115.67	*acre-inches*
38.47	*inches/acre*

Process wastewater applied	
3,160,000.00	*gallons*
116.37	*acre-inches*
2.12	*inches/acre*

Total harvests for the crop	
1	*harvests*

V-7 - 11/03/2022: Oats, silage-soft dough

Field name: V-7 Crop: Oats, silage-soft dough Plant date: 11/03/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	222.05	30.87	233.71	1,834.95
Fresh water	0.25	0.00	0.00	102.71
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	229.30	30.87	233.71	1,937.65
Anticipated crop nutrient removal	160.00	25.60	132.80	0.00
Actual crop nutrient removal	169.48	30.75	190.93	843.81
Nutrient balance	59.83	0.12	42.78	1,093.85
Applied to removed ratio	1.35	1.00	1.22	2.30

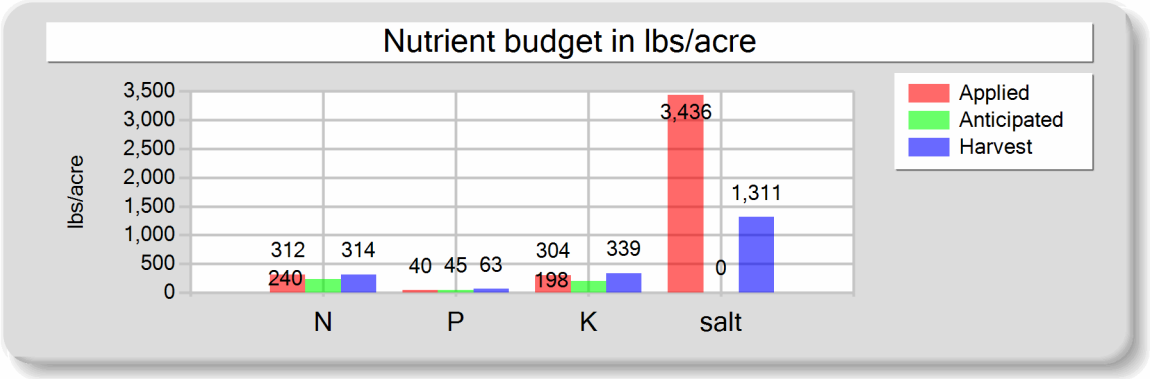
Fresh water applied	
10,506,600.00	*gallons*
386.92	*acre-inches*
11.05	*inches/acre*

Process wastewater applied	
1,300,000.00	*gallons*
47.87	*acre-inches*
1.37	*inches/acre*

Total harvests for the crop	
1	*harvests*

V-7 - 06/26/2023: Corn, silage

Field name: V-7 Crop: Corn, silage Plant date: 06/26/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	304.40	40.02	304.12	3,068.53
Fresh water	0.90	0.00	0.00	367.71
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	312.29	40.02	304.12	3,436.24
Anticipated crop nutrient removal	240.00	45.00	198.00	0.00
Actual crop nutrient removal	314.30	63.31	339.17	1,311.46
Nutrient balance	-2.01	-23.29	-35.05	2,124.78
Applied to removed ratio	0.99	0.63	0.90	2.62

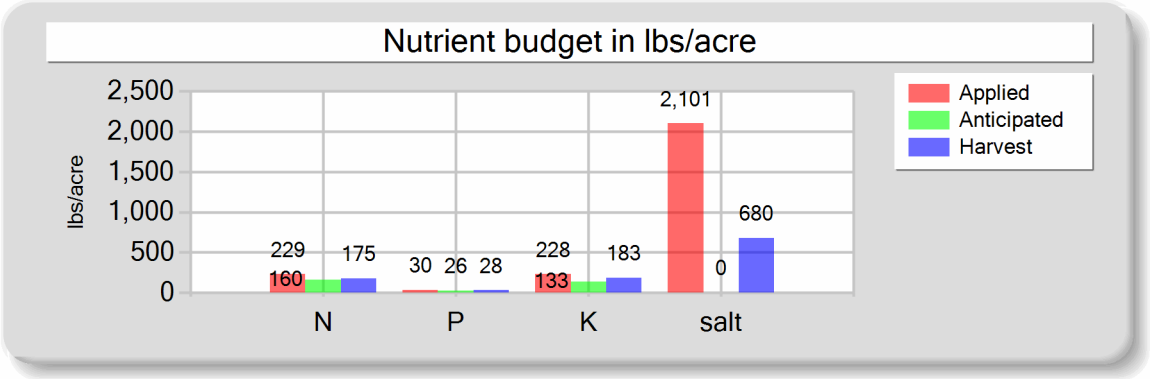
Fresh water applied	
37,614,975.00	*gallons*
1,385.23	*acre-inches*
39.58	*inches/acre*

Process wastewater applied	
2,290,000.00	*gallons*
84.33	*acre-inches*
2.41	*inches/acre*

Total harvests for the crop	
1	*harvests*

V-8 - 11/03/2022: Oats, silage-soft dough

Field name: V-8 Crop: Oats, silage-soft dough Plant date: 11/03/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	221.69	30.10	228.20	2,006.69
Fresh water	0.23	0.00	0.00	94.09
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	228.92	30.10	228.20	2,100.79
Anticipated crop nutrient removal	160.00	25.60	132.80	0.00
Actual crop nutrient removal	174.79	28.48	182.56	679.73
Nutrient balance	54.14	1.61	45.64	1,421.06
Applied to removed ratio	1.31	1.06	1.25	3.09

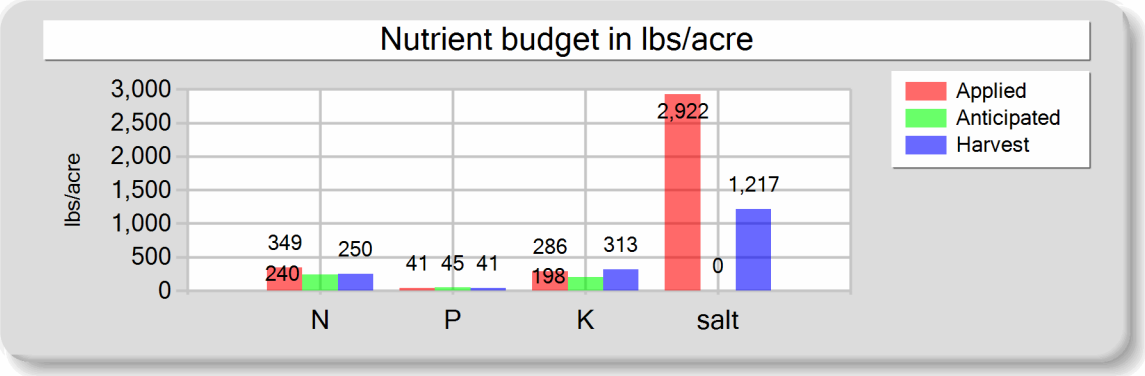
Fresh water applied	
8,250,375.00	*gallons*
303.83	*acre-inches*
10.13	*inches/acre*

Process wastewater applied	
1,250,000.00	*gallons*
46.03	*acre-inches*
1.53	*inches/acre*

Total harvests for the crop	
1	*harvests*

V-8 - 06/26/2023: Corn, silage

Field name: V-8 Crop: Corn, silage Plant date: 06/26/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	340.91	41.07	286.32	2,556.69
Fresh water	0.89	0.00	0.00	364.85
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	348.80	41.07	286.32	2,921.54
Anticipated crop nutrient removal	240.00	45.00	198.00	0.00
Actual crop nutrient removal	249.50	41.24	313.43	1,216.59
Nutrient balance	99.29	-0.17	-27.10	1,704.95
Applied to removed ratio	1.40	1.00	0.91	2.40

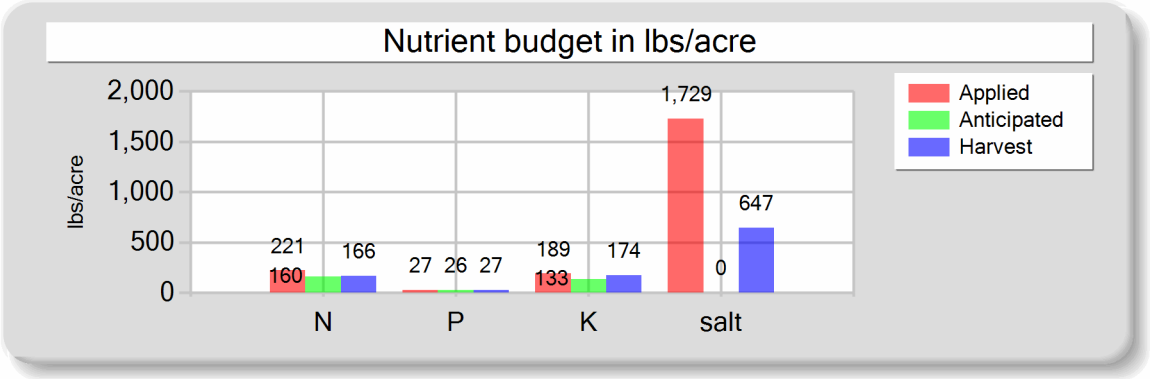
Fresh water applied	
31,991,250.00	*gallons*
1,178.13	*acre-inches*
39.27	*inches/acre*

Process wastewater applied	
1,780,000.00	*gallons*
65.55	*acre-inches*
2.19	*inches/acre*

Total harvests for the crop	
1	*harvests*

V-9 - 11/03/2022: Oats, silage-soft dough

Field name: V-9 Crop: Oats, silage-soft dough Plant date: 11/03/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	214.19	26.57	188.66	1,637.87
Fresh water	0.22	0.00	0.00	91.02
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	221.42	26.57	188.66	1,728.89
Anticipated crop nutrient removal	160.00	25.60	132.80	0.00
Actual crop nutrient removal	166.34	27.11	173.73	646.88
Nutrient balance	55.07	-0.54	14.93	1,082.01
Applied to removed ratio	1.33	0.98	1.09	2.67

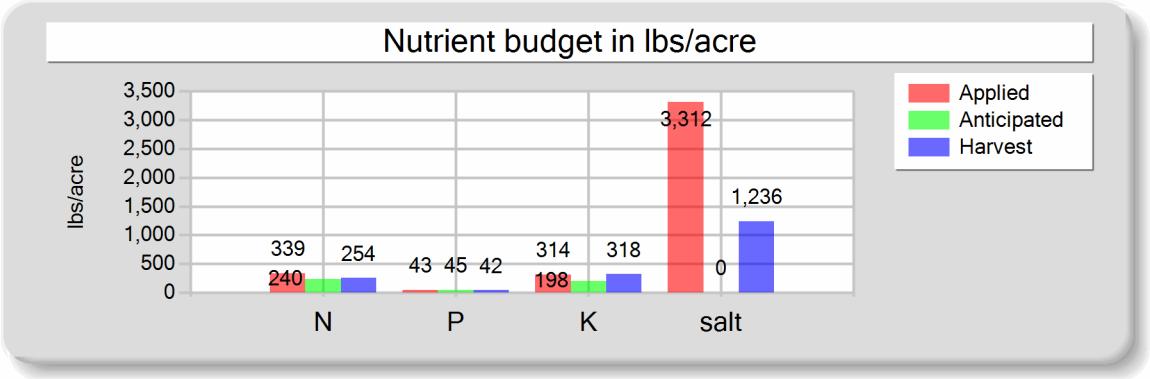
Fresh water applied	
2,660,325.00	*gallons*
97.97	*acre-inches*
9.80	*inches/acre*

Process wastewater applied	
370,000.00	*gallons*
13.63	*acre-inches*
1.36	*inches/acre*

Total harvests for the crop	
1	*harvests*

V-9 - 06/26/2023: Corn, silage

Field name: V-9 Crop: Corn, silage Plant date: 06/26/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	331.33	42.60	314.45	2,941.95
Fresh water	0.90	0.00	0.00	369.85
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	339.23	42.60	314.45	3,311.79
Anticipated crop nutrient removal	240.00	45.00	198.00	0.00
Actual crop nutrient removal	253.52	41.90	318.47	1,236.17
Nutrient balance	85.71	0.70	-4.02	2,075.62
Applied to removed ratio	1.34	1.02	0.99	2.68

Fresh water applied	
10,809,675.00	*gallons*
398.08	*acre-inches*
39.81	*inches/acre*

Process wastewater applied	
640,000.00	*gallons*
23.57	*acre-inches*
2.36	*inches/acre*

Total harvests for the crop	
1	*harvests*

NUTRIENT ANALYSES

A. MANURE ANALYSES

Solid Manure

Sample and source description: Solid Manure

Sample date: 05/24/2023 Material type: Corral solids Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 11.1 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Calcium (mg/kg)	Magnesium (mg/kg)	Sodium (mg/kg)	Sulfur (mg/kg)	Chloride (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	4,000.00	1,600.00	5,600.00							
DL	100.00	100.00	100.00							

Solid Manure

Sample and source description: Solid Manure

Sample date: 10/16/2023 Material type: Corral solids Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 17.2 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Calcium (mg/kg)	Magnesium (mg/kg)	Sodium (mg/kg)	Sulfur (mg/kg)	Chloride (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	5,100.00	2,000.00	5,500.00							
DL	100.00	100.00	100.00							

B. PROCESS WASTEWATER ANALYSES

Lagoon Water

Sample and source description: Lagoon Water

Sample date: 02/20/2023 Material type: Process wastewater Source of analysis: Lab analysis pH: 7.50

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	678.00	466.00	0.00	1.90	77.80	524.00								10,500.00	4,930
DL	1.00	0.50	0.50	0.10	0.10	0.50								10.00	10

Waste water

Sample and source description: Waste water

Sample date: 06/14/2023 Material type: Process wastewater Source of analysis: Lab analysis pH: 7.50

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	716.00	375.00	0.00	0.40	99.60	754.00								13,100.00	5,920
DL	1.00	0.50	0.50	0.10	0.10	0.50								10.00	10

Lagoon Water

Sample and source description: Lagoon Water

Sample date: 09/25/2023 Material type: Process wastewater Source of analysis: Lab analysis pH: 7.50

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	98.70	73.50	0.00	2.70	21.10	92.80								4,210.00	2,280
DL	1.00	0.50	0.50	0.10	0.10	0.50								10.00	10

Lagoon Water

Sample and source description: Lagoon Water

Sample date: 11/15/2023 Material type: Process wastewater Source of analysis: Lab analysis pH: 7.50

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	553.00	407.00	0.00	4.50	73.30	557.00								10,800.00	5,620
DL	1.00	0.50	0.10	0.10	0.10	0.50								10.00	10

C. FRESH WATER ANALYSES

Domestic Well - Houses

Domestic Well - Houses

Houses

Sample description: Houses

Sample date: 11/13/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	0.40	0.00	0.40								4,040.00	0
DL	0.10	1.00	0.10								10.00	1

Domestic Well - Shop

Shop

Sample description: Shop

Sample date: 11/13/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	0.60	0.00	0.60								4,070.00	0
DL	0.10	0.10	0.10								10.00	1

River water

Irrigation water - river water

Sample description: Irrigation water - river water

Sample date: 07/19/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	0.10	0.00	0.10								46.50	41
DL	0.10	1.00	0.10								10.00	10

D. SOIL ANALYSES

V-1

V-1

Sample and source description: V-1

Sample date: 10/11/2023 Source of analysis: Lab analysis

	Nitrate-N (mg/kg)	Total P (mg/kg)	Soluble P (mg/kg)	K (mg/kg)	EC (µmhos/cm)	Organic matter (%)	Total salt (mg/kg)
Value			369.00				
DL			1.00				

V-2

V-2

Sample and source description: V-2

Sample date: 10/31/2023 Source of analysis: Lab analysis

	Nitrate-N (mg/kg)	Total P (mg/kg)	Soluble P (mg/kg)	K (mg/kg)	EC (µmhos/cm)	Organic matter (%)	Total salt (mg/kg)
Value			3,209.00				
DL			1.00				

V-3

V-3

Sample and source description: V-3

Sample date: 10/31/2023 Source of analysis: Lab analysis

	Nitrate-N (mg/kg)	Total P (mg/kg)	Soluble P (mg/kg)	K (mg/kg)	EC (µmhos/cm)	Organic matter (%)	Total salt (mg/kg)
Value			2,098.00				
DL			1.00				

V-4

V-4

V-4

Sample and source description: V-4

Sample date: 10/31/2023 Source of analysis: Lab analysis

	Nitrate-N (mg/kg)	Total P (mg/kg)	Soluble P (mg/kg)	K (mg/kg)	EC (µmhos/cm)	Organic matter (%)	Total salt (mg/kg)
Value			4,101.00				
DL			1.00				

V-5

V-5

Sample and source description: V-5

Sample date: 10/31/2023 Source of analysis: Lab analysis

	Nitrate-N (mg/kg)	Total P (mg/kg)	Soluble P (mg/kg)	K (mg/kg)	EC (µmhos/cm)	Organic matter (%)	Total salt (mg/kg)
Value			1,352.00				
DL			1.00				

V-6

V-6

Sample and source description: V-6

Sample date: 10/31/2023 Source of analysis: Lab analysis

	Nitrate-N (mg/kg)	Total P (mg/kg)	Soluble P (mg/kg)	K (mg/kg)	EC (µmhos/cm)	Organic matter (%)	Total salt (mg/kg)
Value			2,012.00				
DL			1.00				

V-7

V-7

V-7

Sample and source description: V-7

Sample date: 09/28/2023 Source of analysis: Lab analysis

	Nitrate-N (mg/kg)	Total P (mg/kg)	Soluble P (mg/kg)	K (mg/kg)	EC (µmhos/cm)	Organic matter (%)	Total salt (mg/kg)
Value			2,555.00				
DL			1.00				

V-8

V-8

Sample and source description: V-8

Sample date: 09/28/2023 Source of analysis: Lab analysis

	Nitrate-N (mg/kg)	Total P (mg/kg)	Soluble P (mg/kg)	K (mg/kg)	EC (µmhos/cm)	Organic matter (%)	Total salt (mg/kg)
Value			2,911.00				
DL			1.00				

V-9

V-9

Sample and source description: V-9

Sample date: 09/28/2023 Source of analysis: Lab analysis

	Nitrate-N (mg/kg)	Total P (mg/kg)	Soluble P (mg/kg)	K (mg/kg)	EC (µmhos/cm)	Organic matter (%)	Total salt (mg/kg)
Value			2,911.00				
DL			1.00				

E. PLANT TISSUE ANALYSES

V-1 - 11/03/2022: Oats, silage-soft dough

V-1 - 11/03/2022: Oats, silage-soft dough

1

Sample and source description: 1

Sample date: 05/10/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 66.4 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	27,400.00	3,800.00	28,400.00		9.90
DL	100.00	100.00	100.00		0.01

V-1 - 06/26/2023: Corn, silage

1 & 2

Sample and source description: 1 & 2

Sample date: 10/19/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 68.4 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	13,100.00	2,400.00	16,900.00		5.80
DL	100.00	100.00	100.00		0.01

V-2 - 11/03/2022: Oats, silage-soft dough

2

Sample and source description: 2

Sample date: 05/10/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 66.4 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	27,400.00	3,800.00	28,400.00		9.90
DL	100.00	100.00	100.00		0.01

V-2 - 06/26/2023: Corn, silage

1 & 2

Sample and source description: 1 & 2

Sample date: 10/19/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 68.4 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	13,100.00	2,400.00	16,900.00		5.80
DL	100.00	100.00	100.00		0.01

V-3 - 11/03/2022: Oats, silage-soft dough

3

Sample and source description: 3

Sample date: 05/10/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 65.8 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	24,800.00	4,800.00	25,600.00		13.90
DL	100.00	100.00	100.00		0.01

V-3 - 06/26/2023: Corn, silage

3 & 4

Sample and source description: 3 & 4

Sample date: 10/19/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 67.5 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	10,800.00	1,700.00	16,500.00		6.30
DL	100.00	100.00	100.00		0.01

V-4 - 11/03/2022: Oats, silage-soft dough

4

Sample and source description: 4

Sample date: 05/10/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 65.8 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	24,800.00	4,800.00	25,600.00		13.90
DL	100.00	100.00	100.00		0.01

V-4 - 06/26/2023: Corn, silage

3 & 4

Sample and source description: 3 & 4

Sample date: 10/19/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 67.5 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	10,800.00	1,700.00	16,500.00		6.30
DL	100.00	100.00	100.00		0.01

V-5 - 11/03/2022: Oats, silage-soft dough

5

Sample and source description: 5

Sample date: 05/10/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 66.4 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	23,200.00	4,100.00	28,600.00		10.80
DL	100.00	100.00	100.00		0.01

V-5 - 06/26/2023: Corn, silage

5

Sample and source description: 5

Sample date: 10/19/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 68.0 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	13,800.00	2,400.00	16,400.00		6.50
DL	100.00	100.00	100.00		0.01

V-6 - 11/03/2022: Oats, silage-soft dough

6A

Sample and source description: 6A

Sample date: 05/10/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 66.3 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	27,100.00	4,400.00	27,700.00		11.40
DL	100.00	100.00	100.00		0.01

6B

Sample and source description: 6B

Sample date: 05/10/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 66.5 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	26,700.00	4,600.00	27,200.00		13.70
DL	100.00	100.00	100.00		0.01

V-6 - 06/26/2023: Corn, silage

V-6 - 06/26/2023: Corn, silage

6A

Sample and source description: 6A

Sample date: 10/19/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 66.3 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	13,800.00	2,200.00	15,000.00		5.60
DL	100.00	100.00	100.00		0.01

6B

Sample and source description: 6B

Sample date: 10/19/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 60.6 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	13,800.00	2,600.00	15,500.00		5.90
DL	100.00	100.00	100.00		0.01

V-7 - 11/03/2022: Oats, silage-soft dough

7

Sample and source description: 7

Sample date: 05/10/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 65.2 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	23,700.00	4,300.00	26,700.00		11.80
DL	100.00	100.00	100.00		0.01

V-7 - 06/26/2023: Corn, silage

V-7 - 06/26/2023: Corn, silage

7

Sample and source description: 7

Sample date: 10/19/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 62.3 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	13,900.00	2,800.00	15,000.00		5.80
DL	100.00	100.00	100.00		0.01

V-8 - 11/03/2022: Oats, silage-soft dough

8

Sample and source description: 8

Sample date: 05/10/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 66.0 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	27,000.00	4,400.00	28,200.00		10.50
DL	100.00	100.00	100.00		0.01

V-8 - 06/26/2023: Corn, silage

8 & 9

Sample and source description: 8 & 9

Sample date: 10/19/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 61.4 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	12,100.00	2,000.00	15,200.00		5.90
DL	100.00	100.00	100.00		0.01

V-9 - 11/03/2022: Oats, silage-soft dough

9

Sample and source description: 9

Sample date: 05/10/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 66.0 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	27,000.00	4,400.00	28,200.00		10.50
DL	100.00	100.00	100.00		0.01

V-9 - 06/26/2023: Corn, silage

8 & 9

Sample and source description: 8 & 9

Sample date: 10/19/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 61.4 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	12,100.00	2,000.00	15,200.00		5.90
DL	100.00	100.00	100.00		0.01

F. SUBSURFACE (TILE) DRAINAGE ANALYSES

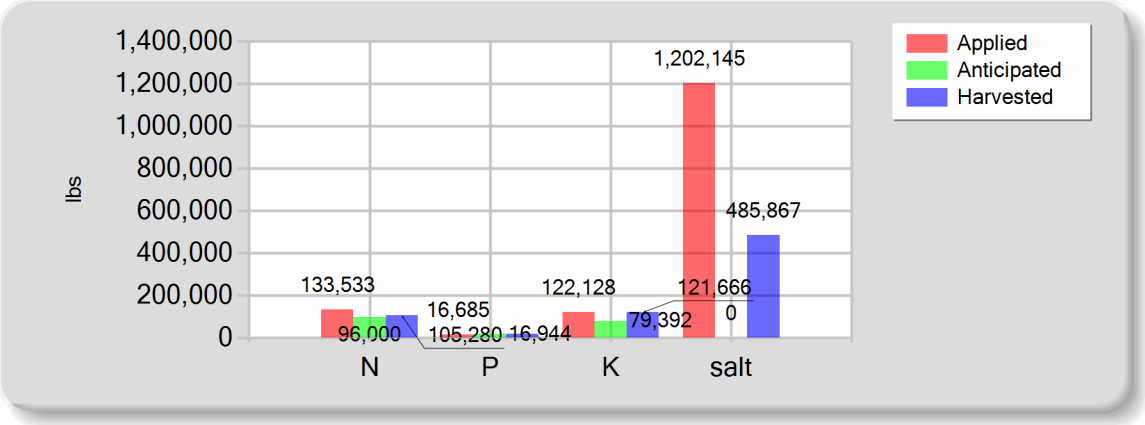
No subsurface (tile) drainage analyses entered.

NUTRIENT APPLICATIONS, POTENTIAL REMOVAL, AND BALANCE

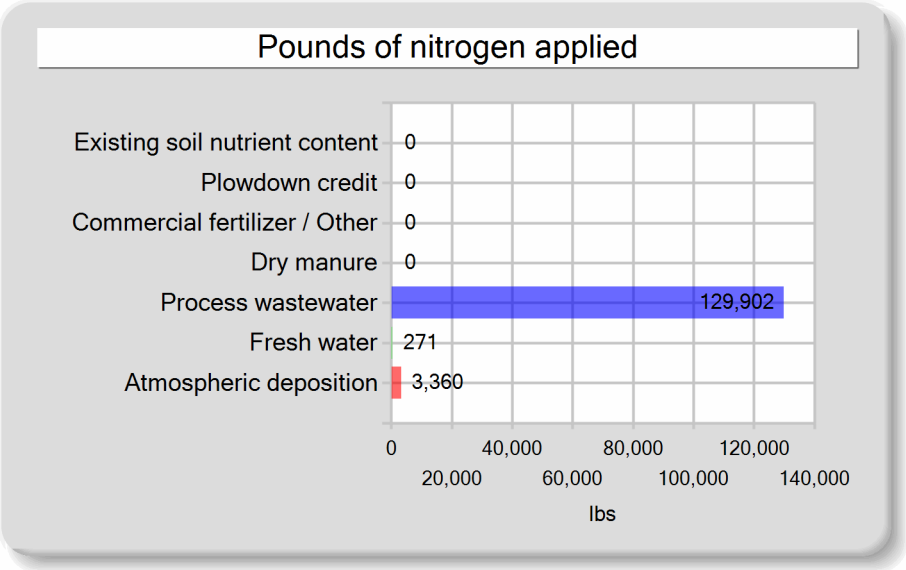
A. SUMMARY OF NUTRIENT APPLICATIONS, POTENTIAL REMOVAL, AND BALANCE

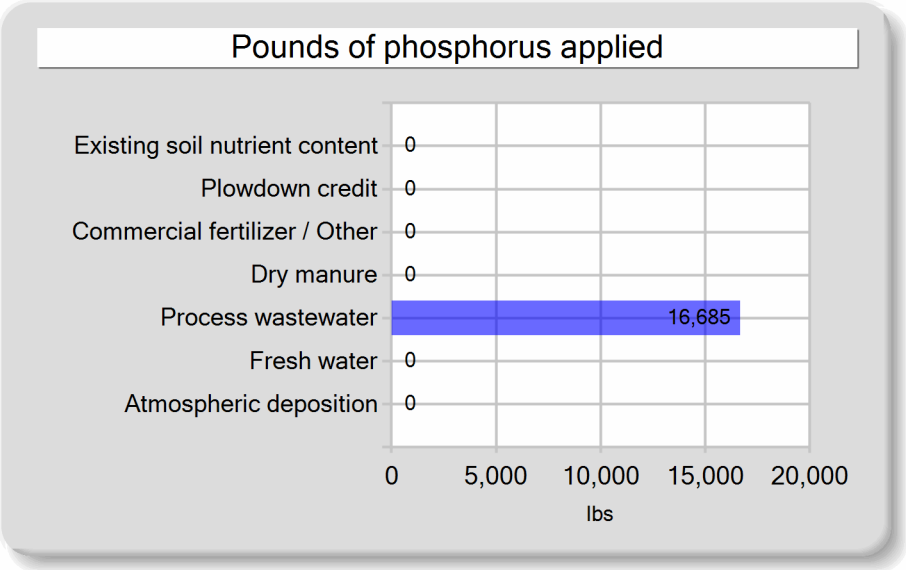
	Total N (lbs)	Total P (lbs)	Total K (lbs)	Total salt (lbs)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	129,901.95	16,685.32	122,128.49	1,091,213.90
Fresh water	270.56	0.00	0.00	110,931.24
Atmospheric deposition	3,360.00	0.00	0.00	0.00
Total nutrients applied	133,532.51	16,685.32	122,128.49	1,202,145.14
Anticipated crop nutrient removal	96,000.00	16,944.00	79,392.00	0.00
Actual crop nutrient removal	105,280.45	18,100.37	121,666.30	485,866.83
Nutrient balance	28,252.07	-1,415.05	462.19	716,278.31
Applied to removed ratio	1.27	0.92	1.00	2.47

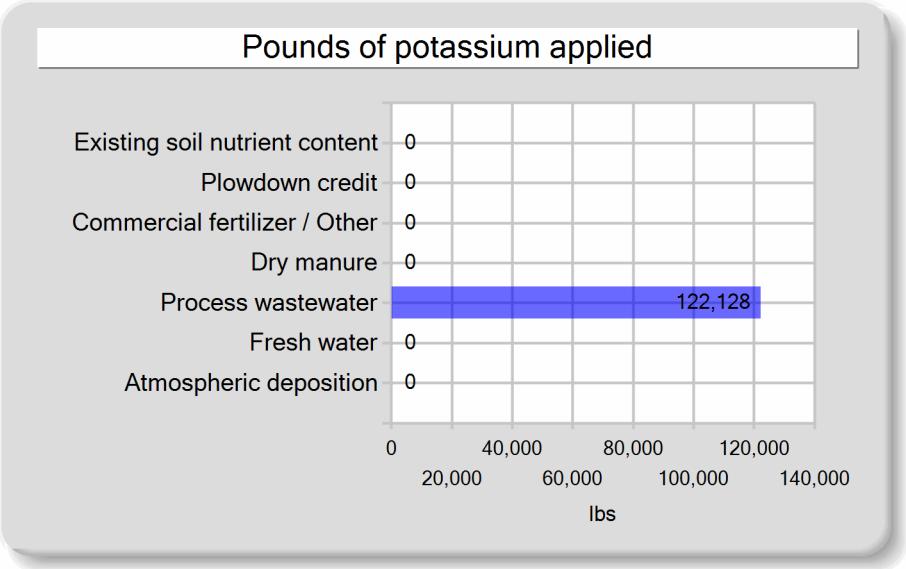
B. POUNDS OF NUTRIENT APPLIED VS. CROP REMOVAL

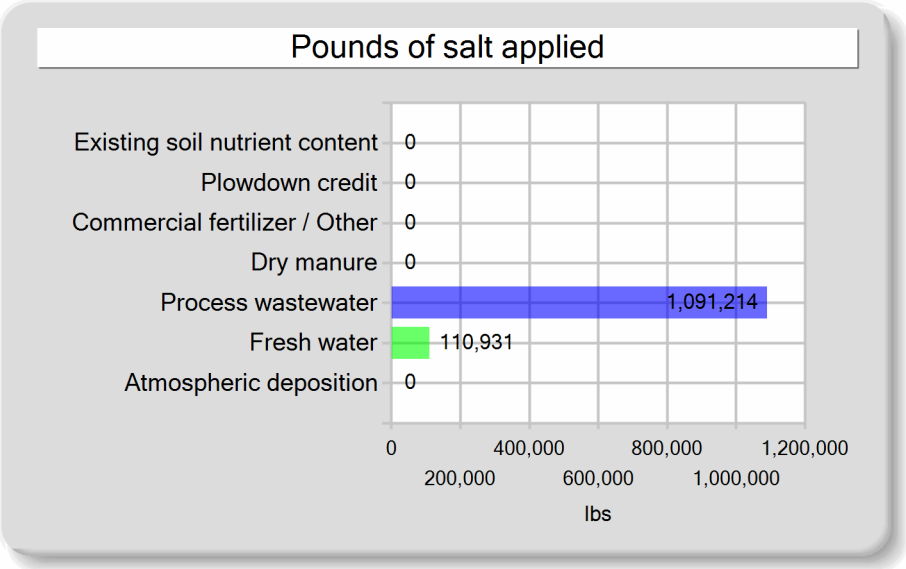


C. POUNDS OF NUTRIENT APPLIED BY MATERIAL TYPE









EXCEPTION REPORTING

A. MANURE, PROCESS WASTEWATER, AND OTHER DAIRY WASTE DISCHARGES

The following is a summary of all manure and process wastewater discharges from the production area to surface water or to land areas (land application areas or otherwise) when not in accordance with the facility's Nutrient Management Plan.

No manure or process wastewater discharges occurred during the reporting period.

B. STORM WATER DISCHARGES

The following is a summary of all storm water discharges from the production area to surface water during the reporting period when not in accordance with the facility 's Nutrient Management Plan.

No stormwater discharges occurred during the reporting period.

C. LAND APPLICATION AREA TO SURFACE WATER DISCHARGES

The following is a summary of all discharges from the land application area to surface water that have occurred during the reporting period when not in accordance with the facility's Nutrient Management Plan.

No land application area to surface water discharges occurred during the reporting period.

NUTRIENT MANAGEMENT PLAN AND EXPORT AGREEMENT STATEMENTS

A. NUTRIENT MANAGEMENT PLAN STATEMENTS

Was the facility's NMP updated in the reporting period?	Yes
Was the facility's NMP developed by a certified nutrient management planner (specialist) as specified in Attachment C of the General Order?	Yes
Was the facility's NMP approved by a certified nutrient management planner (specialist) as specified in Attachment C of the General Order?	Yes

B. EXPORT AGREEMENT STATEMENT

Are there any written agreements with third parties to receive manure or process wastewater that are new or were revised within the reporting period?	No

ADDITIONAL NOTES

A. NOTES

*SOILS were reported as P2O5**

CERTIFICATION

A. OWNER AND/OR OPERATOR CERTIFICATION

I certify under penalty of law that I have personally examined and am familiar with the information submitted in this document and all attachments and that, based on my inquiry of those individuals immediately responsible for obtaining the information, I believe that the information is true, accurate, and complete. I am aware that there are significant penalties for submitting false information, including the possibility of fine and imprisonment.

Michael Brasil (Jun 19, 2024 20:37 PDT)	Jonathan Vitoria (Jun 19, 2024 11:09 PDT)
SIGNATURE OF OWNER OF FACILITY	SIGNATURE OF OPERATOR OF FACILITY
Michael Brasil	Jonathan Vitoria
PRINT OR TYPE NAME	PRINT OR TYPE NAME
06/19/2024	06/19/2024
DATE	DATE

DELLAVALLE™
LABORATORY INC

R & J Avila Inc
8119 Ellington Ave
Hilmar, CA 95324

Account# 00-0019840
Account Manager: Ben Nydam
Submitted By: Rui & John Avila
Ranch: Vitoria Dairy

Received: 11/14/2023 15:45
Reported: 11/21/2023 09:16

Samples in this Report

Lab ID	Sample	Matrix	Sampled By	Crop	Date Sampled
23K0667-01	Houses	Ag Water	T. Madruga		11/13/2023 15:00
23K0667-02	Shop	Ag Water	T. Madruga		11/13/2023 16:00

Default Cooler Temperature on Receipt °C: 4.5
Containers Intact
COC/Labels Agree
Received On Ice

Notes and Definitions

Item	Definition
H	Hold Time Exceeded
MCL	Drinking Water Maximum Contaminant Level
ND	Analyte NOT DETECTED at or above the reporting limit.
NES	Not Enough Sample
*	Not Taken
RPD	Relative Percent Difference
%REC	Percent Recovery
Source	Sample that was matrix spiked or duplicated.

Scott McFriedland

Laboratory Director/Technical Manager

ELAP Certification #1595
A2LA Certification #6440.02

The results in this report apply to the samples as received and were analyzed in accordance with the chain of custody document. This analytical report must be reproduced in its entirety. All results listed in this report are for the exclusive use of the submitting party. Dellavalle Laboratory, Inc. assumes no responsibility for report alteration, separation, detachment or third party interpretation.

DELLAVALLE™
LABORATORY INC

R & J Avila Inc
8119 Ellington Ave
Hilmar, CA 95324

Account# 00-0019840
Account Manager: Ben Nydam
Submitted By: Rui & John Avila
Ranch: Vitoria Dairy

Received: 11/14/2023 15:45
Reported: 11/21/2023 09:16

Sample Results

Sample: Houses
23K0667-01 (Water)

Sampled: 11/13/2023 15:00
Sampled By: T. Madruga

Analyte	Result	Units	Reporting Limit	DIL	DW MCL	Date/Time Analyzed	Method	Notes	Batch
Electrical Conductivity	**4.04**	mmhos/cm	0.01	1		11/16/23 12:36	SM 2510 B		BEK0477
Electrical Conductivity umhos	**4040**	umhos/cm	10.0	1		11/16/23 12:36	SM 2510 B		BEK0477
Ammonia (as N)	ND	mg/L	0.00	1		11/13/23 15:00	Field		BEK0415
Nitrate Nitrogen as NO3N	**0.4**	mg/L	0.1	1	10	11/14/23 19:22	EPA 300.0		BEK0412
Temperature	**25.0**	units	0.0	1		11/16/23 12:36	SM 4500-H+	H	BEK0477
pH	**7.6**	units	1.0	1		11/16/23 12:36	SM 4500-H+	H	BEK0477

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DELLAVALLE™
LABORATORY INC

R & J Avila Inc
8119 Ellington Ave
Hilmar, CA 95324

Account# 00-0019840
Account Manager: Ben Nydam
Submitted By: Rui & John Avila
Ranch: Vitoria Dairy

Received: 11/14/2023 15:45
Reported: 11/21/2023 09:16

Sample Results
(Continued)

Sample: Shop
23K0667-02 (Water)

Sampled: 11/13/2023 16:00
Sampled By: T. Madruga

Analyte	Result	Units	Reporting Limit	DIL	DW MCL	Date/Time Analyzed	Method	Notes	Batch
Electrical Conductivity	**4.07**	mmhos/cm	0.01	1		11/16/23 12:37	SM 2510 B		BEK0477
Electrical Conductivity umhos	**4070**	umhos/cm	10.0	1		11/16/23 12:37	SM 2510 B		BEK0477
Ammonia (as N)	ND	mg/L	0.00	1		11/13/23 16:00	Field		BEK0415
Nitrate Nitrogen as NO3N	**0.6**	mg/L	0.1	1	10	11/14/23 19:42	EPA 300.0		BEK0412
Temperature	**25.0**	units	0.0	1		11/16/23 12:37	SM 4500-H+	H	BEK0477
pH	**7.8**	units	1.0	1		11/16/23 12:37	SM 4500-H+	H	BEK0477

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Quality Control

Analyte	Result	Qual	Reporting Limit	Units	Spike Level	Source Result	%REC	%REC Limits	RPD	RPD Limit
Batch: BEK0412										
Blank (BEK0412-BLK1)					Prepared & Analyzed: 11/14/2023					
Nitrate Nitrogen as NO3N	ND		0.1	mg/L						
Blank (BEK0412-BLK2)					Prepared & Analyzed: 11/14/2023					
Nitrate Nitrogen as NO3N	ND		0.1	mg/L						
Blank (BEK0412-BLK3)					Prepared: 11/14/2023 Analyzed: 11/15/2023					
Nitrate Nitrogen as NO3N	ND		0.1	mg/L						
LCS (BEK0412-BS1)					Prepared & Analyzed: 11/14/2023					
Nitrate Nitrogen as NO3N	5.0		0.1	mg/L	5.000		100	90-110		
LCS (BEK0412-BS2)					Prepared: 11/14/2023 Analyzed: 11/15/2023					
Nitrate Nitrogen as NO3N	5.0		0.1	mg/L	5.000		100	90-110		
Duplicate (BEK0412-DUP1)		**Source: 23K0676-01**			Prepared & Analyzed: 11/14/2023					
Nitrate Nitrogen as NO3N	15.7		0.1	mg/L		15.5			0.802	10
Duplicate (BEK0412-DUP2)		**Source: 23K0676-02**			Prepared: 11/14/2023 Analyzed: 11/15/2023					
Nitrate Nitrogen as NO3N	36.4		0.1	mg/L		36.4			0.0192	10
Matrix Spike (BEK0412-MS1)		**Source: 23K0676-01**			Prepared & Analyzed: 11/14/2023					
Nitrate Nitrogen as NO3N	20.6		0.1	mg/L	5.000	15.5	102	90-110		
Matrix Spike (BEK0412-MS2)		**Source: 23K0676-02**			Prepared: 11/14/2023 Analyzed: 11/15/2023					
Nitrate Nitrogen as NO3N	42.1		0.1	mg/L	5.000	36.4	113	90-110		
Reference (BEK0412-SRM1)					Prepared & Analyzed: 11/14/2023					
Nitrate Nitrogen as NO3N	9.8			mg/L	10.00		97.9	90-110		
Reference (BEK0412-SRM2)					Prepared: 11/14/2023 Analyzed: 11/15/2023					
Nitrate Nitrogen as NO3N	9.9			mg/L	10.00		98.7	90-110		
Reference (BEK0412-SRM3)					Prepared: 11/14/2023 Analyzed: 11/15/2023					
Nitrate Nitrogen as NO3N	9.9			mg/L	10.00		99.2	90-110		

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Quality Control
(Continued)

Analyte	Result	Qual	Reporting Limit	Units	Spike Level	Source Result	%REC	%REC Limits	RPD	RPD Limit
Batch: BEK0477										
Blank (BEK0477-BLK1)					Prepared: 11/15/2023	Analyzed: 11/16/2023				
Temperature	25.0		0.0	units						
Electrical Conductivity	ND		0.01	mmhos/cm						
pH	5.5		1.0	units						
Electrical Conductivity umhos	ND		10.0	umhos/cm						
Blank (BEK0477-BLK2)					Prepared: 11/15/2023	Analyzed: 11/16/2023				
Electrical Conductivity	ND		0.01	mmhos/cm						
Temperature	25.0		0.0	units						
Electrical Conductivity umhos	ND		10.0	umhos/cm						
pH	7.5		1.0	units						
Blank (BEK0477-BLK3)					Prepared: 11/15/2023	Analyzed: 11/16/2023				
Electrical Conductivity	ND		0.01	mmhos/cm						
Temperature	25.0		0.0	units						
pH	7.3		1.0	units						
Electrical Conductivity umhos	ND		10.0	umhos/cm						
Duplicate (BEK0477-DUP1)		**Source: 23K0667-01**			Prepared: 11/15/2023	Analyzed: 11/16/2023				
Electrical Conductivity	4.04		0.01	mmhos/cm		4.04			0.0495	10
pH	7.8		1.0	units		7.6			1.82	10
Electrical Conductivity umhos	4040		10.0	umhos/cm		4040			0.0495	10
Duplicate (BEK0477-DUP2)		**Source: 23K0676-03**			Prepared: 11/15/2023	Analyzed: 11/16/2023				
Electrical Conductivity	1.35		0.01	mmhos/cm		1.36			0.791	10
Electrical Conductivity umhos	1350		10.0	umhos/cm		1360			0.791	10
pH	7.6		1.0	units		7.6			0.264	10
Reference (BEK0477-SRM1)					Prepared: 11/15/2023	Analyzed: 11/16/2023				
Electrical Conductivity	578			umhos/cm	538.0		107	90-110		
Reference (BEK0477-SRM2)					Prepared: 11/15/2023	Analyzed: 11/16/2023				
pH	7.5			units	7.520		100	67021-101.32		
Reference (BEK0477-SRM3)					Prepared: 11/15/2023	Analyzed: 11/16/2023				
Electrical Conductivity	1060			umhos/cm	1000		106	90-110		
Electrical Conductivity umhos	1060			umhos/cm	1000		106	90-110		
Reference (BEK0477-SRM4)					Prepared: 11/15/2023	Analyzed: 11/16/2023				
Electrical Conductivity	1080			umhos/cm	1000		108	90-110		
Electrical Conductivity umhos	1080			umhos/cm	1000		108	90-110		
Reference (BEK0477-SRM5)					Prepared: 11/15/2023	Analyzed: 11/16/2023				
Electrical Conductivity	1060			umhos/cm	1000		106	90-110		

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Quality Control
(Continued)

Analyte	Result	Qual	Reporting Limit	Units	Spike Level	Source Result	%REC	%REC Limits	RPD	RPD Limit
Batch: BEK0477 (Continued)										
Reference (BEK0477-SRM5)					Prepared: 11/15/2023	Analyzed: 11/16/2023				
Electrical Conductivity umhos	1060			umhos/cm	1000		106	90-110		
Reference (BEK0477-SRM6)					Prepared: 11/15/2023	Analyzed: 11/16/2023				
pH	4.0			units	4.000		100	97.5-102.5		
Reference (BEK0477-SRM7)					Prepared: 11/15/2023	Analyzed: 11/16/2023				
pH	4.0			units	4.000		100	97.5-102.5		
Reference (BEK0477-SRM8)					Prepared: 11/15/2023	Analyzed: 11/16/2023				
pH	4.0			units	4.000		100	97.5-102.5		

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11/14/23 15:45 23K0667

JG

DELLAVALLE LABORATORY, INC.

1910 W. McKinley Avenue, Suite 110 • Fresno, CA 93728
www.dellavallelab.com 559 233-6129 · 800 228-9896 · Fax 559 268-8174

No. Samples: 2 No. of Bottles: 2

Purchase Order No | Bill To: | Acct # 19840 | Cons # 08

Results Need By

Water Type: [ ] Drinking Water [ ] Wastewater
[X] Ag Water [ ] Groundwater [ ] Monitoring Well

Name: R & J Avila Inc
Address: 8119 Ellington Avenue
City: Hilmar **State:** CA **Zip:** 95324
Telephone: (209) 669-7919 **Fax:** (209) 634-4604
Cell/Email: 209 678-3086 Rui / 209 678-1723 John frontdesk.randjavila@gmail.com

COPY TO:

REQUESTED BY: Rui & John Avila
PROJECT Vitoria Dairy
ID CROP: N/A

[X] Copy of Chain [X] QA/QC Documents
Sampled By: T. Madruga

Other:

Analysis and Bottles Required: (Please indicate Analysis)

(X) DWW1: EC, pH, NO_3-N (NH4-N Field Test)
(1-1 Liter Plastic, Unpreserved) White Per Sample
( ) DWW2: DWW1 Plus SO_4, CO_3, HCO_3, Cl, Ca, Mg, Na, TDS
(1-1 Liter Plastic, Unpreserved) White Per Sample
( ) DCW1: EC, NO_3-N, TDS + TKN & TN
(1-1 Liter Plastic, Unpreserved) White Per Sample
( ) DPW1: EC, pH, NO_3-N, NH_4-N, TKN, TDS, TP, TK
(1-1 Liter Plastic, Unpreserved) White Per Sample
( ) DPW2: DPW1 Plus SO_4, CO_3, HCO_3, Cl, Ca, Mg, Na
(1-1 Liter Plastic, Unpreserved) White Per Sample
( ) Other

	Description of Samples	Date Sampled	Time Sampled	Rec'd Temp °C	Field NH_4-N
1	Houses	11/13/2023	1500	4.5	ND
2	Shop	11/13/2023	1600	4.7	ND
3					
4					
5					
6					
7					
8					
9					
10					

IR Thermometer SN: 192603727
Correction Factor: 0°C
Calibration Due: 12/22/2023
Location: Laboratory

CHAIN OF CUSTODY

Carrier	Signature	Company	Received (Date/Time)	Relinquished (Date/Time)
First	John Avila	R & J Avila, Inc.	11/13/2023 @ 1600	11/14/2023 @ 900
Second	JLP	Dellavalle Laboratory, Inc.	11/14/2023 @ 900	11-14-23 3:30
Third				
Fourth			11/14 15:45	

I guarantee that as the client, or on behalf of client named, I have the authority to contract the above requested services. Should it be found that I do not have such authority, I agree to be personally liable for all costs and, if there should be action against me for this breach, reasonable attorneys' fees. It is understood that payment is expected to be cash with samples unless terms have been previously arranged. Terms are net 30 days; overdue accounts will be charged a liquidated damage fee of 2% per month (annually 24%) or $5.00 per month whichever is greater.
If payment is not made when due and a legitimate dispute exists concerning the product or services of Dellavalle Laboratory, Inc., it will be submitted to mediation under the Rules and Procedures of Creative Alternative to Litigation, Inc. (*cal*). If the dispute is not resolved in mediation, then the dispute will be submitted to binding arbitration through ***cal*** under its Rules and Procedures. The parties will equally bear the costs of mediation/arbitration. If, however, the mediator declares that no legitimate dispute exists, then debtor will pay all mediation and arbitration costs, and in the event of arbitration,reasonable attorneys' fees of Dellavalle Laboratory.

Invoicing Information: ***Shipping***
Sampling hrs $ In
Miles $ Out
Consulting

Amt Paid	Rec By	Check #	Date

Signature

Samples recieved in cooler with ice (coolant)
[ ] Yes [ ] No

11/14/23 15:45 23K0667

Shipping Information: Fed Ex ☐ UPS ☐ GSO ☐ Picked-Up ☒ Walk In ☐ DLI Sampler ☐
Ontrac ☐ PMS ☐ USPS ☐ Other ☐ _____

Shipping Container: Ice Chest ☒ None ☐ Box ☐

Refrigerant: Wet Ice ☒ Blue Ice: ☐ None: ☐

Container(s) Received	1	2	3	4	5	6	7	8	9	10
Plastics:										
100 mL sterile plastic $Na_2S_2O_3$ (Green)										
250 mL unpreserved (White)										
250 mL HNO_3 (Red)										
250 mL H_2SO_4 (Yellow)										
500 mL unpreserved (White)										
500 mL HNO_3 (Red)										
500 mL H_2SO_4 (Yellow)										
1 L unpreserved (White)	1	1								
1 L unpreserved (BOD) (Purple)										
1 L HNO_3 (Red)										
1 L H_2SO_4 (Yellow)										
VOA Vials in sets of 3:										
40 mL VOA, HCl (Blue)										
40 mL VOA, $Na_2S_2O_3$ (Green)										
40 mL VOA, unpreserved (White)										
Glass:										
250 mL AG unpreserved (White)										
250 mL AG H_2SO_4 (Yellow)										
250 mL AG $Na_2S_2O_3$ (Green)										
250 mL AG $Na_2S_2O_3$ + MCAA										
500 mL glass unpreserved (White)										
500 mL AG HCl (Blue)										
1 L AG unpreserved (White)										
1 L AG H_2SO_4 (Yellow)										
1 L AG $Na_2S_2O_3$ (Green)										
1 L AG HCl (Blue)										
Special:										
Cr^{6+} w/Borate/HCO3/CO3										
Cyanide - 500 mL NaOH										
Asbestos - 2 - 1L P wrapped in foil										
500 mL or 1 L **AP** $Na_2S_2O_3$ (Green)										
1 L **AG** or **P** NaOH + ZnAc (Sulfide)										
Chlorite/Bromate 250 mL AG with EDA										
TTHM										
HAA5										
40 mL VOA, $Na_2S_2O_3$ + MCAA										